Tarea_3_DML_Benjamín_Araneda_Wolf

July 5, 2025

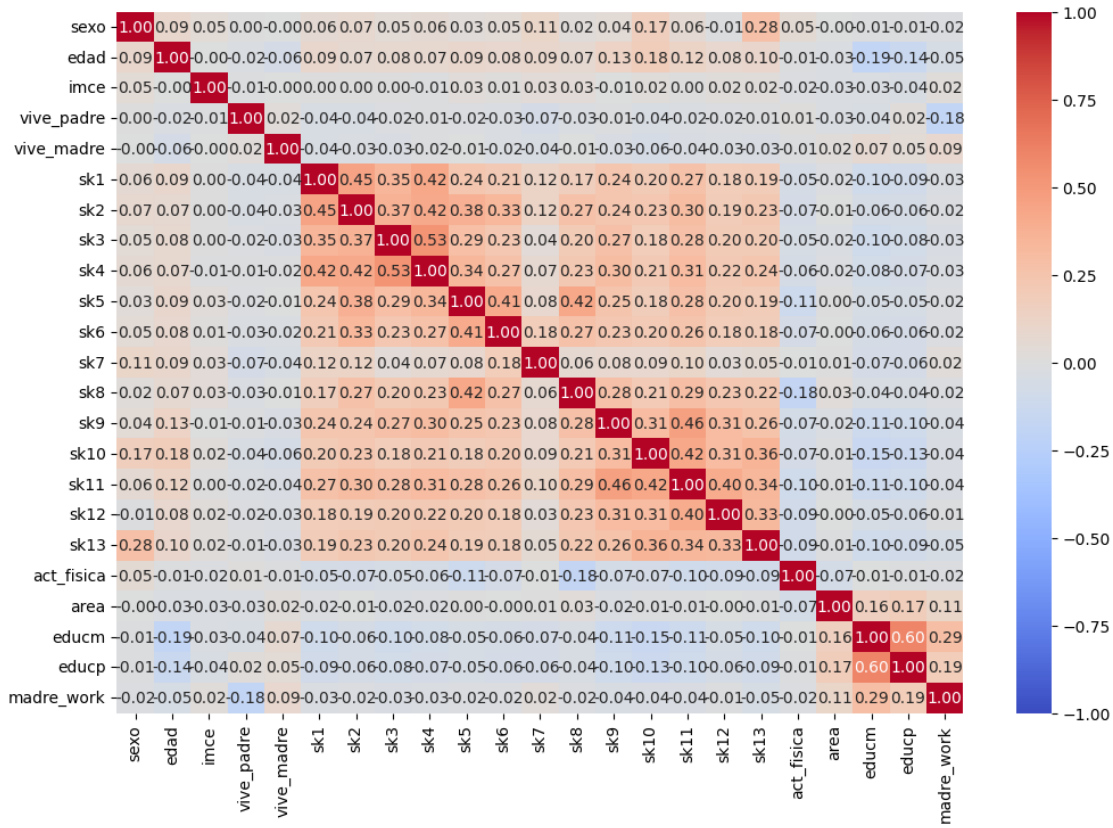
1 VARIABLES DICCIONARIO

- sexo: sexo del estudiante (1: mujer)
- edad: edad del estudiante (meses)
- imce: indice de masa corporal estandarizado (puntaje z)
- vive_padre: 1 si el padre vive en el hogar (2 indica que tiene mas de un padre)
- vive_madre: 2(**es 1, modificación**) si la madre vive en el hogar (2 indica que tiene mas de una madre)
- area: urbana=1, rural=0
- sk1: muestra afecto a padres (1: siempre - 5: nunca)
- sk2: muestra afecto a sus pares (1: siempre - 5: nunca)
- sk3: expresa sus sentimientos (1: siempre - 5: nunca)
- sk4: usa gestos para mostrar sentimientos (1: siempre - 5: nunca)
- sk5: juega con otros (1: siempre - 5: nunca)
- sk6: comparte sus cosas con otros (1: siempre - 5: nunca)
- sk7: es agresivo (1: siempre - 5: nunca)
- sk8: participa en juegos grupales (1: siempre - 5: nunca)
- sk9: hace preguntas a adultos (1: siempre - 5: nunca)
- sk10: tiene interes por libros (1: siempre - 5: nunca)
- sk11: tiene interes por su entorno (1: siempre - 5: nunca)
- sk12: juega a armar y desarmar cosas (1: siempre - 5: nunca)
- sk13: tiene expresiones artisticas (1: siempre - 5: nunca)
- act_fisica: frecuencia actividad fisica (1: nunca - 5: 5 o mas veces a la semana)
- educm: años de escolaridad de la madre
- educp: años de escolaridad del padre
- madre_work: si la madre trabaja (-1: labor domestica, 0: desempleada, 1: empleada)

	sexo	edad	imce	vive_padre	vive_madre	sk1	sk2	sk3	sk4	sk5	…	\
0	1	85	0.75	1	1	1	1	1	2	1	…	
1	0	76	0.71	0	1	1	1	1	1	1	…	
2	1	68	0.27	0	1	2	2	3	2	1	…	
3	1	84	2.05	1	1	1	1	1	1	1	…	
4	0	86	1.05	1	1	1	1	1	1	1	…	
…	…	…	…	…	…	…	…	…	…	…	…	
31050	0	78	1.63	1	1	1	3	1	2	2	…	
31051	1	79	2.57	1	1	1	1	2	2	1	…	
31052	0	78	2.12	1	1	1	1	1	1	1	…	
31053	1	78	-0.43	1	1	1	1	1	1	2	…	
31054	0	78	-0.55	0	1	2	2	1	1	1	…	

	sk9	sk10	sk11	sk12	sk13	act_fisica	area	educm	educp	madre_work
0	2	2	2	3	2	NaN	0	11.0	11	-1
1	1	1	1	1	1	5.0	0	8.0	8	1
2	2	3	2	1	3	NaN	1	13.0	13	1
3	1	1	1	1	1	2.0	1	16.0	12	-1
4	1	1	1	1	1	1.0	1	17.0	15	0
…	…	…	…	…	…	…	…	…	…	…
31050	2	2	2	1	1	2.0	1	13.0	13	-1
31051	1	3	2	1	4	3.0	1	18.0	19	0
31052	1	3	1	1	1	3.0	1	13.0	9	1
31053	1	2	1	1	2	2.0	1	17.0	15	1
31054	1	1	1	1	1	1.0	1	18.0	11	1

[31055 rows x 23 columns]



2 revisamos valores anormales

```
array([1, 0])
```

```
62 107
```

```
8.916666666666666
```

```
vive_madre
1    29903
0     1096
2       56
Name: count, dtype: int64
```

```
vive_padre
1    22354
0     8693
2        8
Name: count, dtype: int64
```

```
area
```

```
1    28308
0     2747
Name: count, dtype: int64



educm
13.0    13180
9.0      2615
15.0     1886
11.0     1807
16.0     1707
18.0     1629
17.0     1601
12.0     1040
10.0      976
14.0      760
20.0      615
22.0      542
7.0       532
8.0       522
6.0       325
5.0       304
19.0      237
4.0       165
3.0       116
0.0        66
2.0        57
Name: count, dtype: int64

 educp
13    13376
9      3089
11     2168
17     1713
15     1523
18     1484
16     1243
10     1080
12     1052
20      855
14      564
7       545
22      543
8       434
19      320
5       317
6       276
```

```
4       186
3       132
0        93
2        62
Name: count, dtype: int64
```

3 revisamos nans

```
sexo             0
edad             0
imce             0
vive_padre       0
vive_madre       0
sk1              0
sk2              0
sk3              0
sk4              0
sk5              0
sk6              0
sk7              0
sk8              0
sk9              0
sk10             0
sk11             0
sk12             0
sk13             0
act_fisica    1004
area             0
educm          373
educp            0
madre_work       0
dtype: int64
```

```
31055
```

```
31055
30051
```

```
29696
```

```
Empty DataFrame
Columns: [sexo, edad, imce, vive_padre, vive_madre, sk1, sk2, sk3, sk4, sk5,
sk6, sk7, sk8, sk9, sk10, sk11, sk12, sk13, act_fisica, area, educm, educp,
madre_work]
Index: []
```

```
[0 rows x 23 columns]

sexo          0
edad          0
imce          0
vive_padre    0
vive_madre    0
sk1           0
sk2           0
sk3           0
sk4           0
sk5           0
sk6           0
sk7           0
sk8           0
sk9           0
sk10          0
sk11          0
sk12          0
sk13          0
act_fisica    0
area          0
educm         0
educp         0
madre_work    0
dtype: int64
```

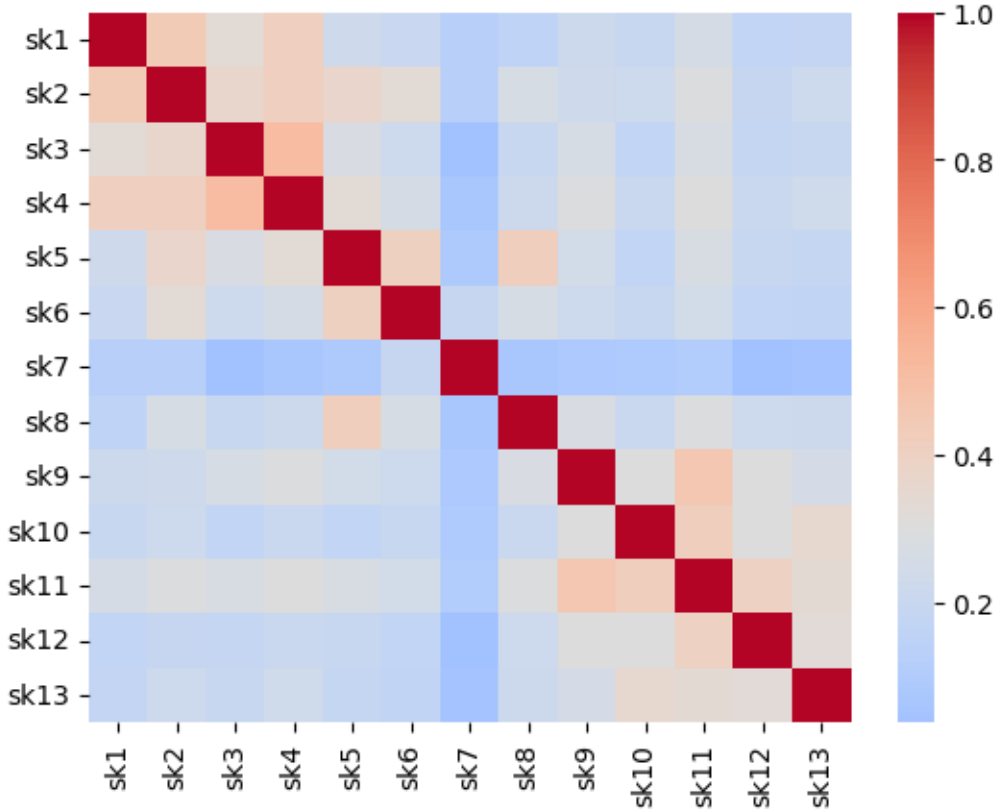
4 revisamos outliers

```
array([1, 2, 4, 3, 5])

       sk1  sk2  sk3  sk4  sk5  sk6  sk7  sk8  sk9  sk10  sk11  sk12  sk13
1        1    1    1    1    1    1    4    1    1     1     1     1     1
3        1    1    1    1    1    1    2    1    1     1     1     1     1
4        1    1    1    1    1    1    3    1    1     1     1     1     1
5        1    2    1    1    1    1    4    1    1     1     1     1     1
6        1    1    1    2    2    2    3    2    3     3     3     2     2
…     …  …  …  …  …  …  …  …  …  …   …   …   …
31050    1    3    1    2    2    1    1    3    2     2     2     1     1
31051    1    1    2    2    1    1    1    2    1     3     2     1     4
31052    1    1    1    1    1    1    1    1    1     3     1     1     1
31053    1    1    1    1    2    2    1    2    1     2     1     1     2
31054    2    2    1    1    1    1    5    1    1     1     1     1     1

[29696 rows x 13 columns]
```

```
<Axes: >
```




```
[0.28732908 0.18595434 0.10025747 0.07968822 0.06627543 0.06442256
 0.05212843 0.04403429 0.03243927 0.02898623 0.02484142 0.02079568
 0.01284758]
```

5 Pregunta 2

```
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
```

```
Index(['sk1', 'sk2', 'sk3', 'sk4', 'sk5', 'sk6', 'sk7', 'sk8', 'sk9', 'sk10',
       'sk11', 'sk12', 'sk13'],
      dtype='object')
```

```
array([[0.171221, 0.540217, 0.151458],
       [0.174149, 0.512531, 0.354330],
       [0.179785, 0.587108, 0.161938],
```

```
        [0.195805, 0.685256, 0.194526],
        [0.121301, 0.214133, 0.716727],
        [0.162524, 0.203313, 0.486711],
        [0.070301, 0.074530, 0.146613],
        [0.269211, 0.097481, 0.478541],
        [0.495887, 0.205273, 0.201636],
        [0.555989, 0.121166, 0.127875],
        [0.665920, 0.206954, 0.202009],
        [0.524127, 0.109526, 0.141501],
        [0.480548, 0.151480, 0.131725]])

        FACTOR 1  FACTOR 2  FACTOR 3
sk1 -> [0.171221 0.540217 0.151458]
VALOR MAXIMO -> 0.5402172337490964 PErtenece al factor -> 2


sk2 -> [0.174149 0.512531 0.354330]
VALOR MAXIMO -> 0.5125312056354531 PErtenece al factor -> 2


sk3 -> [0.179785 0.587108 0.161938]
VALOR MAXIMO -> 0.5871075837335075 PErtenece al factor -> 2


sk4 -> [0.195805 0.685256 0.194526]
VALOR MAXIMO -> 0.6852555692945741 PErtenece al factor -> 2


sk5 -> [0.121301 0.214133 0.716727]
VALOR MAXIMO -> 0.7167265444773256 PErtenece al factor -> 3


sk6 -> [0.162524 0.203313 0.486711]
VALOR MAXIMO -> 0.48671058090988484 PErtenece al factor -> 3


sk7 -> [0.070301 0.074530 0.146613]
VALOR MAXIMO -> 0.14661347127040691 PErtenece al factor -> 3


sk8 -> [0.269211 0.097481 0.478541]
VALOR MAXIMO -> 0.4785411203226863 PErtenece al factor -> 3


sk9 -> [0.495887 0.205273 0.201636]
VALOR MAXIMO -> 0.4958873723760414 PErtenece al factor -> 1
```

```
sk10 -> [0.555989 0.121166 0.127875]
VALOR MAXIMO -> 0.555989262013226 PErtenece al factor -> 1


sk11 -> [0.665920 0.206954 0.202009]
VALOR MAXIMO -> 0.6659196336192474 PErtenece al factor -> 1


sk12 -> [0.524127 0.109526 0.141501]
VALOR MAXIMO -> 0.5241272731903086 PErtenece al factor -> 1


sk13 -> [0.480548 0.151480 0.131725]
VALOR MAXIMO -> 0.48054774844542103 PErtenece al factor -> 1

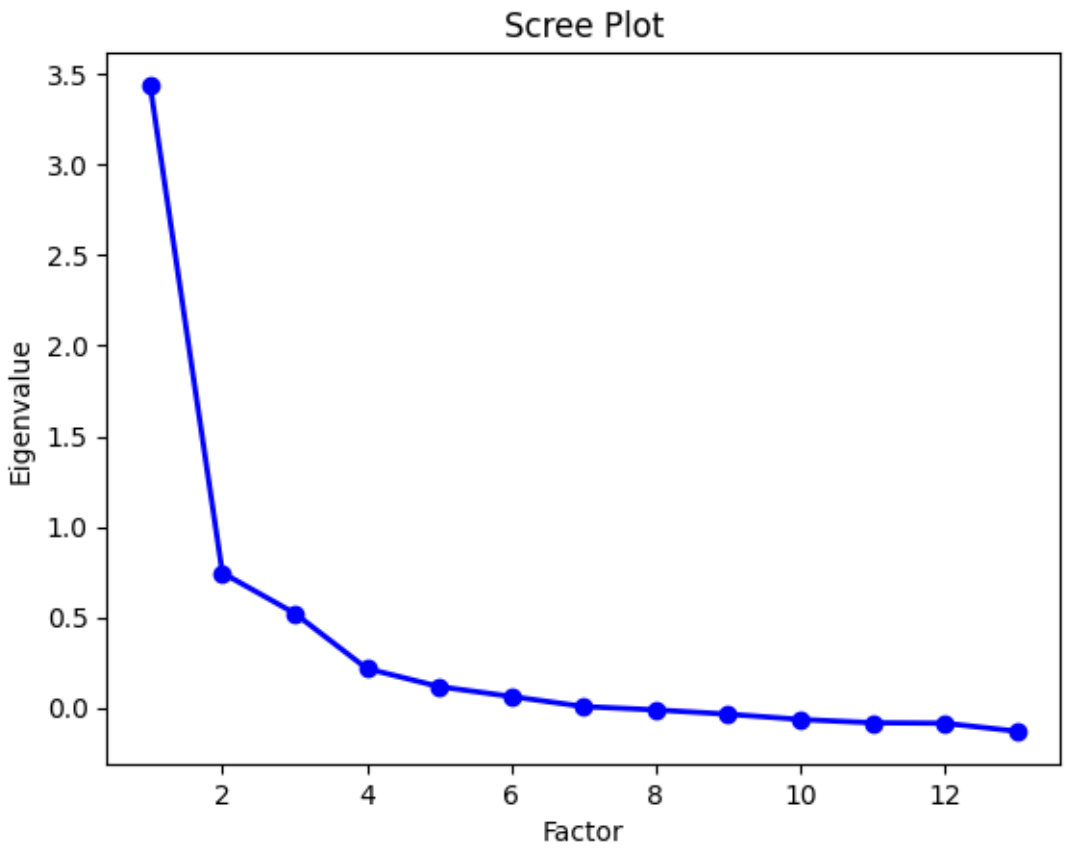


Index(['sk1', 'sk2', 'sk3', 'sk4', 'sk5', 'sk6', 'sk7', 'sk8', 'sk9', 'sk10',
       'sk11', 'sk12', 'sk13'],
      dtype='object')

13

(array([4.034140, 1.350905, 1.114858, 1.001343, 0.791246, 0.707033,
        0.701430, 0.690630, 0.608183, 0.529379, 0.517172, 0.491397,
        0.462285]),
 array([3.438847, 0.745863, 0.522710, 0.217186, 0.117540, 0.062923,
        0.007478, -0.011080, -0.034611, -0.063772, -0.082158, -0.084646,
        -0.128859]))

array([0.344091, 0.418566, 0.403242, 0.545755, 0.574264, 0.304637,
       0.031992, 0.310979, 0.328698, 0.340157, 0.527086, 0.306728,
       0.271224])
```



6 Respuesta pregunta 2

7 variables y factores (se determinaron 3 factores)

- sk1 -> Factor 2
- sk2 -> Factor 2
- sk3 -> Factor 2
- sk4 -> Factor 2
- sk5 -> Factor 3
- sk6 -> Factor 3
- sk7 -> Factor 3
- sk8 -> Factor 3
- sk9 -> Factor 1
- sk10 -> Factor 1
- sk11 -> Factor 1
- sk12 -> Factor 1
- sk13 -> Factor 1

se obtuvo lo anterior en cuanto a factores - 5 variables al factor 1 (sk 13 / 12 / 11/ 10/ 9) - 4 vaariables al factor 2 (sk 8 / 7 / 6/ 5) - 4 variables al factor 3 (sk 4 / 3 / 2/ 1)

Cabe recalcar que sk7 es sospechoso de no ser una variable informativa, pues si, tiene asignado el factor 3, mas su coeficiente es bastante bajo en comparación al resto, y por supuesto, los demas valores aun más bajos, por lo que, mas que decir que pertenece al factor 3, se podria decir a cuales factores pertenece menos, y por "descarte" se le asigno uno(el 3). Al menos de lo que arroja este analisis.

Respeco a la decision de 3 factores: se eligieron 3 puesto que el incremento a partir del 4 factor es marginal en cuanto a variabilidad explicada, ademas notemos que en output de comunalities, podemos observar como la variable sk7(su varianza) es muy poco explicada por los factores, lo que re afirma el hecho de su firmeza con respecto a su pertenencia a uno de estos

- nota: Curiosamente pareciera que el que decidio tomar estas variables, penso en algun aspecto, coloco todo lo que creyo importante, y luego en otro aspecto y asi, o quizas no, pero es curioso que tengan cierto orden…

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13])

         0         1         2         3         4         5         6   \
0  4.034140  1.350905  1.114858  1.001343  0.791246  0.707033  0.701430
1  3.438847  0.745863  0.522710  0.217186  0.117540  0.062923  0.007478

        7         8         9         10        11        12
0  0.69063  0.608183  0.529379  0.517172  0.491397  0.462285
1 -0.01108 -0.034611 -0.063772 -0.082158 -0.084646 -0.128859

(array([1.752965, 1.605630, 1.348825]),
 array([0.134843, 0.123510, 0.103756]),
 array([0.134843, 0.258353, 0.362109]))
```

8 Pregunta 3

```
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(

array([[0.171221, 0.540217, 0.151458],
       [0.174149, 0.512531, 0.354330],
       [0.179785, 0.587108, 0.161938],
       [0.195805, 0.685256, 0.194526],
       [0.121301, 0.214133, 0.716727],
       [0.162524, 0.203313, 0.486711],
       [0.070301, 0.074530, 0.146613],
       [0.269211, 0.097481, 0.478541],
       [0.495887, 0.205273, 0.201636],
       [0.555989, 0.121166, 0.127875],
       [0.665920, 0.206954, 0.202009],
       [0.524127, 0.109526, 0.141501],
```

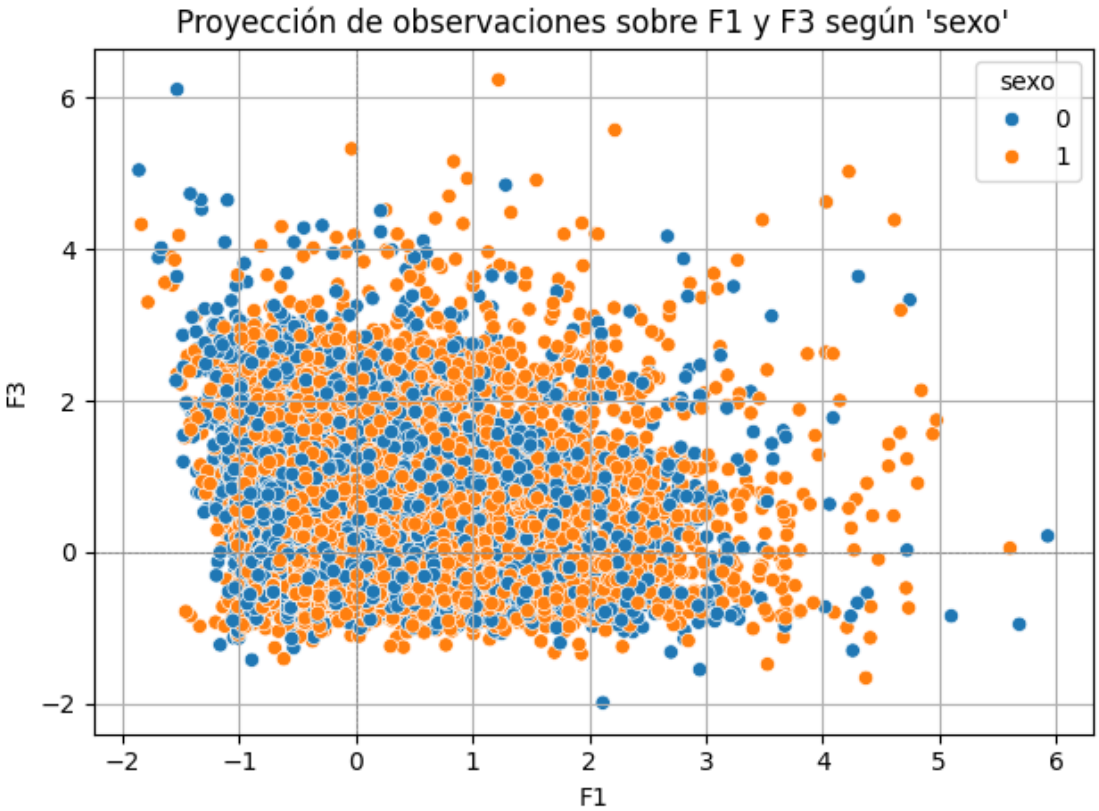
```
       [0.480548, 0.151480, 0.131725]])
```

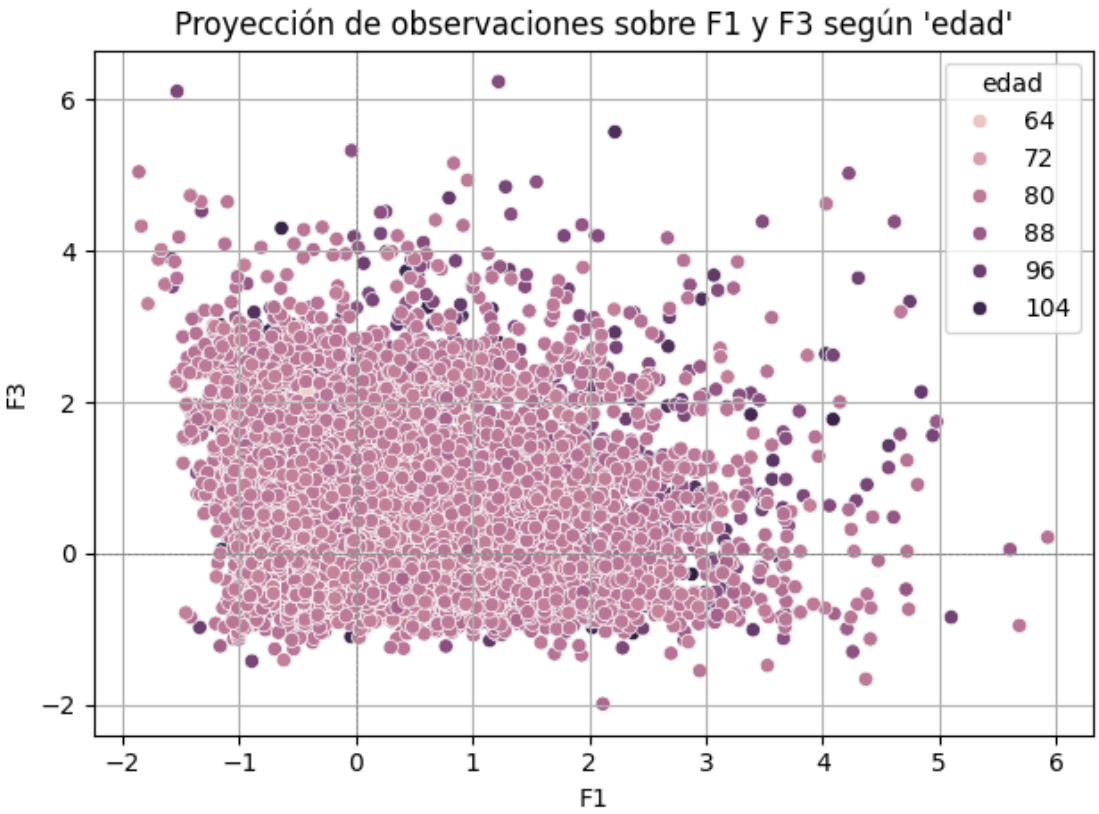
```
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
```

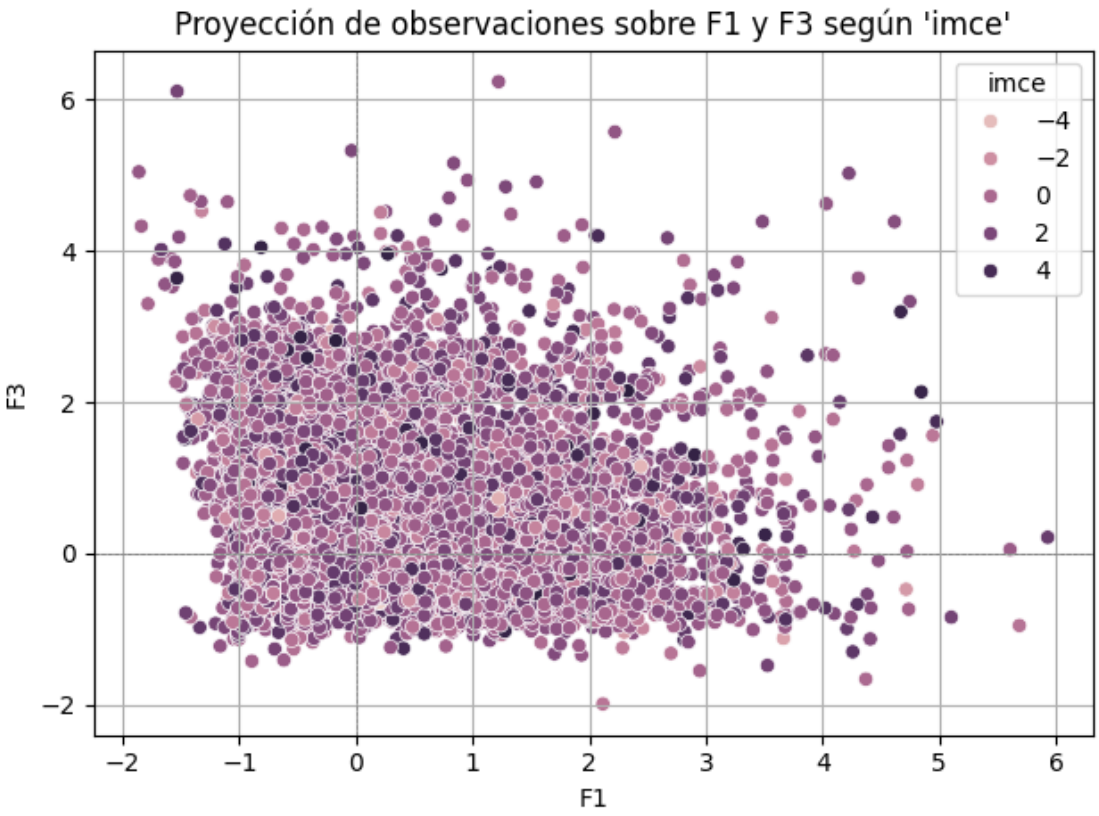
```
Index(['sexo', 'edad', 'imce', 'vive_padre', 'vive_madre', 'sk1', 'sk2', 'sk3',
       'sk4', 'sk5', 'sk6', 'sk7', 'sk8', 'sk9', 'sk10', 'sk11', 'sk12',
       'sk13', 'act_fisica', 'area', 'educm', 'educp', 'madre_work'],
      dtype='object')
```

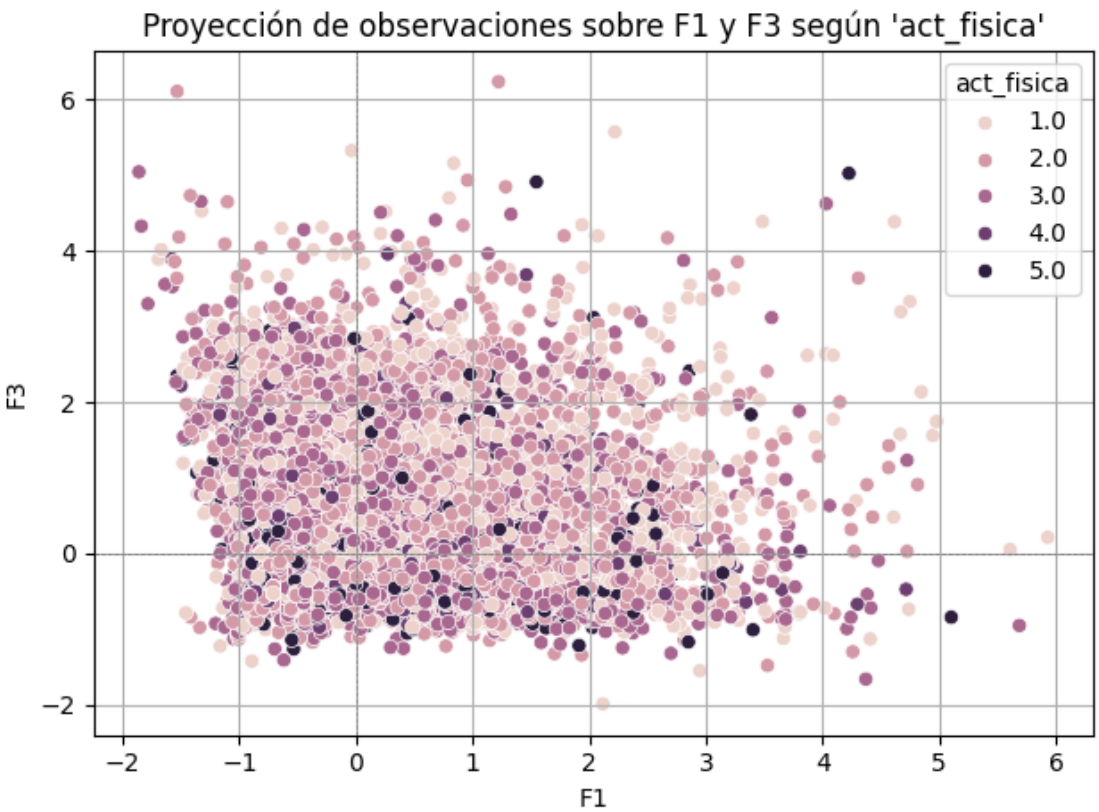
#graficamos todo

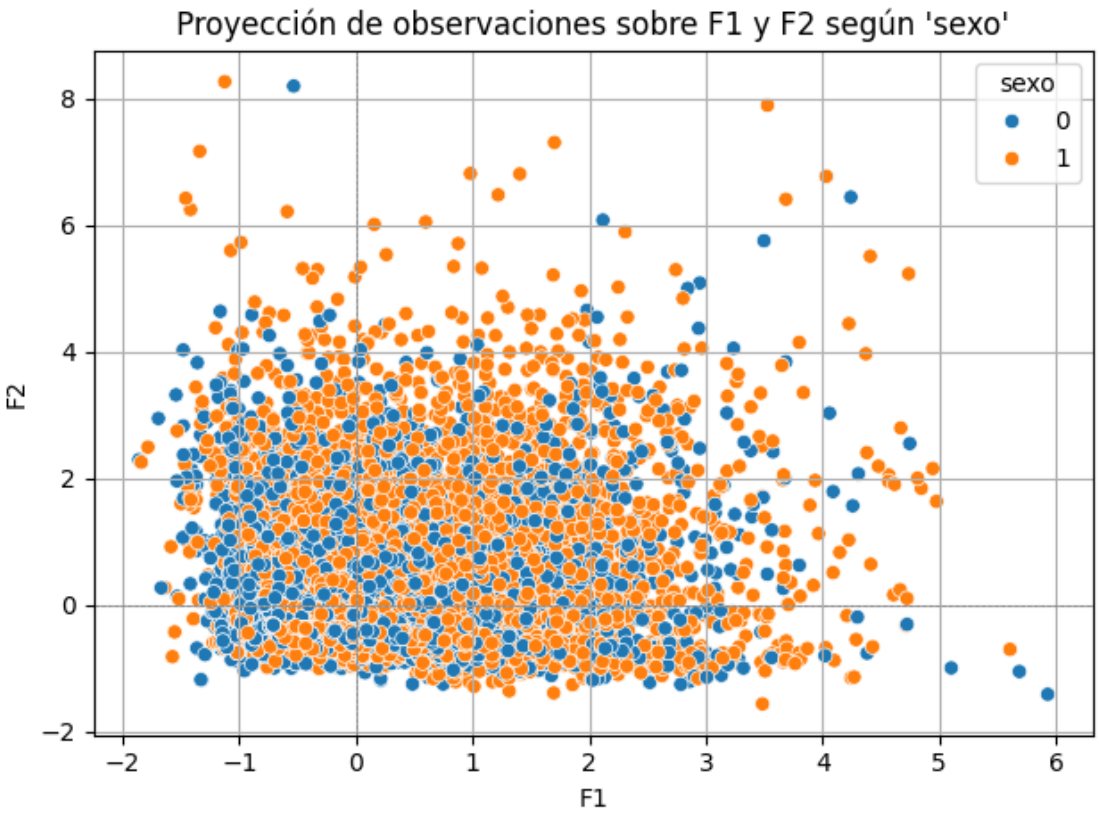
combinaciones= (F1,F3) (F1,F2) (F2,F3)

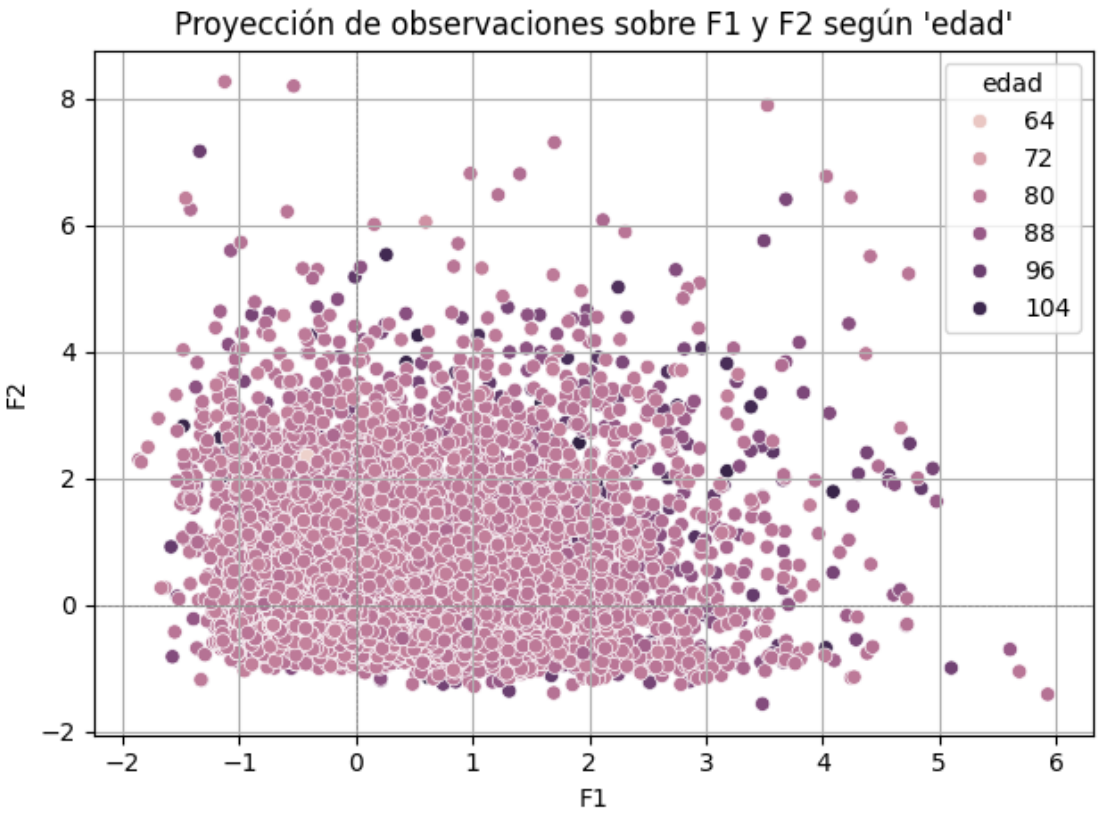


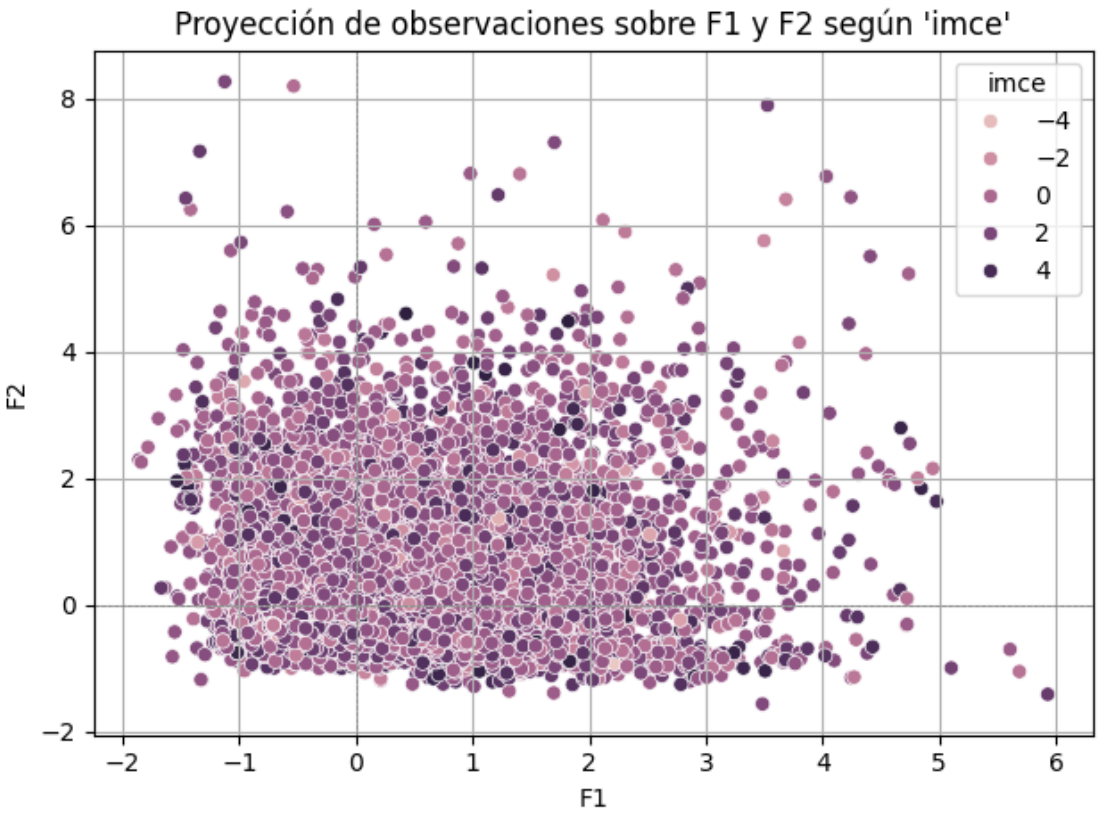

Proyección de observaciones sobre F1 y F3 según 'edad'
edad
64
72
80
88
96
104
F3
F1
6
4
2
0
−2
−2
−1
0
1
2
3
4
5
6

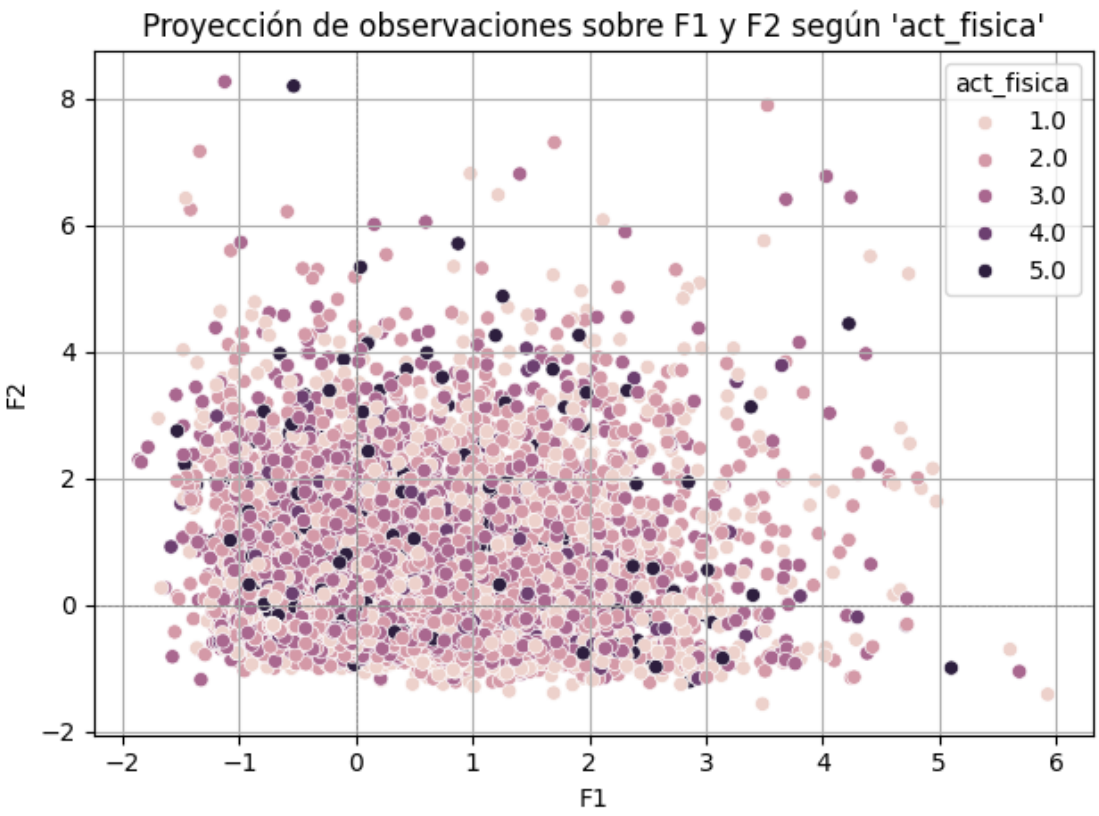

Proyección de observaciones sobre F1 y F3 según 'imce'
imce
−4
−2
0
2
4
F3
F1
6
4
2
0
−2
−2
−1
0
1
2
3
4
5
6

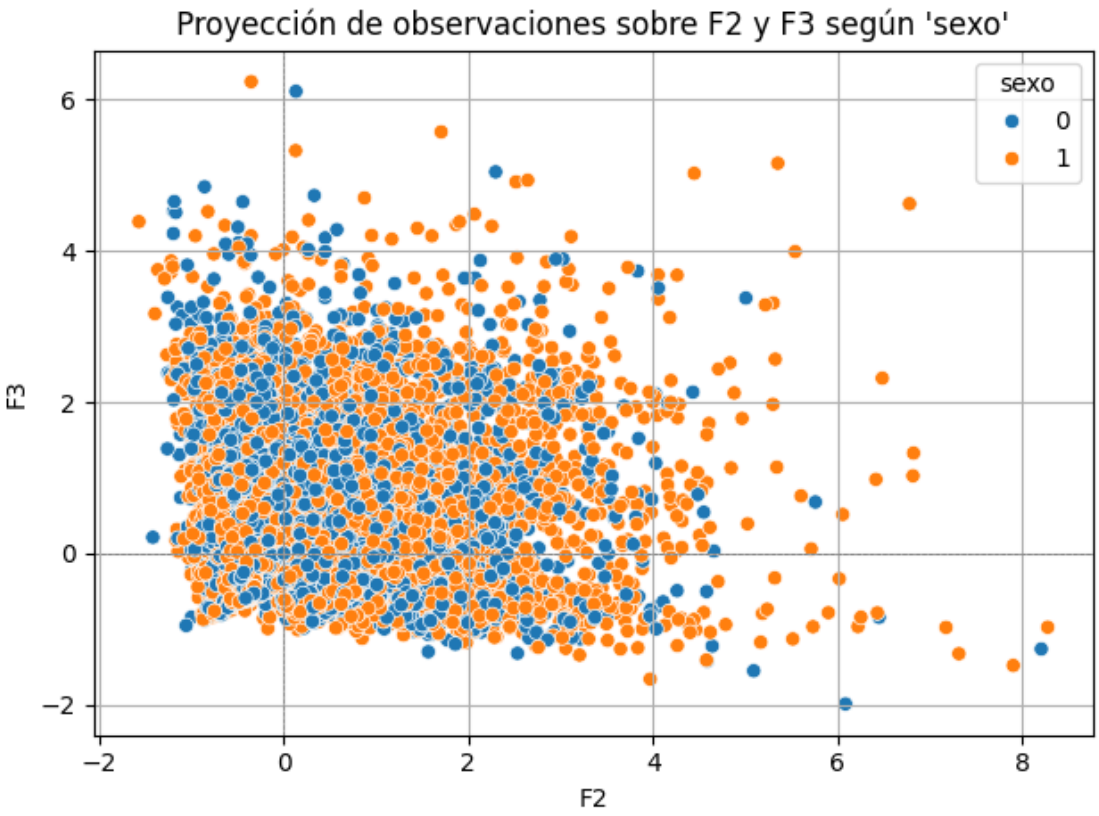

Proyección de observaciones sobre F1 y F3 según 'act_fisica'
act_fisica
1.0
2.0
3.0
4.0
5.0
F3
F1
6
4
2
0
−2
−2
−1
0
1
2
3
4
5
6

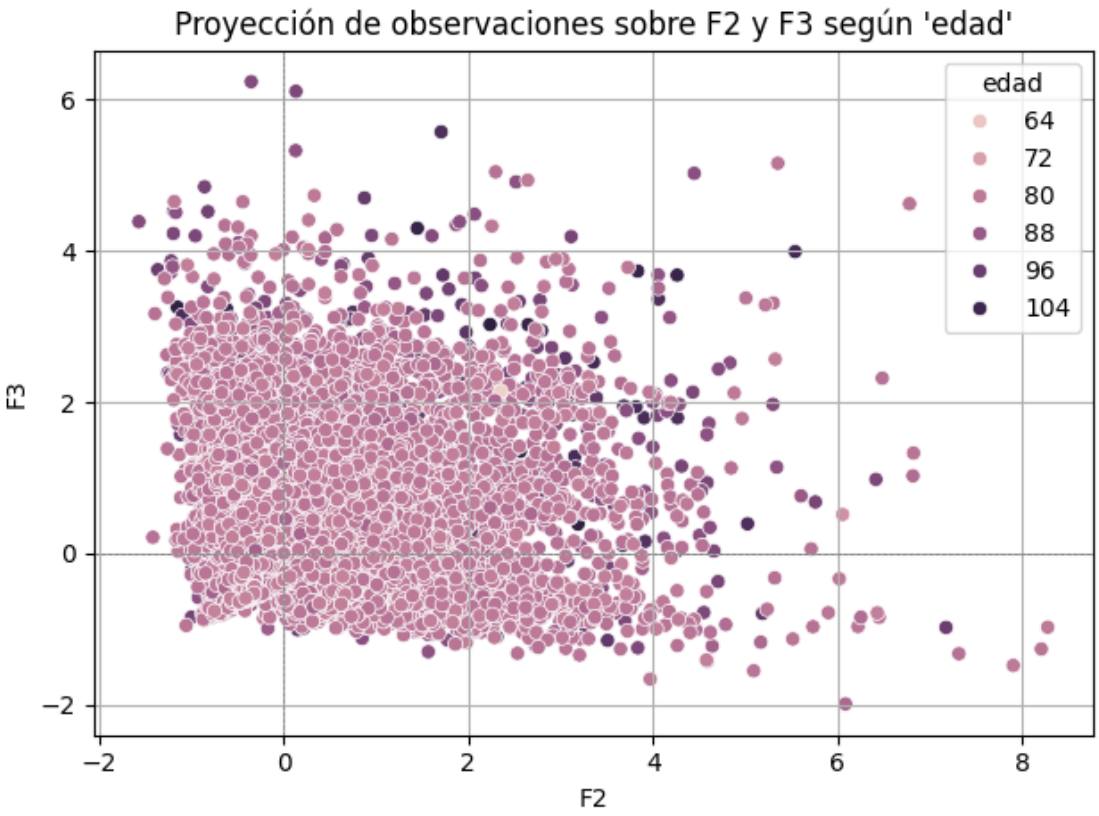

Proyección de observaciones sobre F1 y F2 según 'sexo'
sexo
0
1
F2
F1
8
6
4
2
0
−2
−2
−1
0
1
2
3
4
5
6

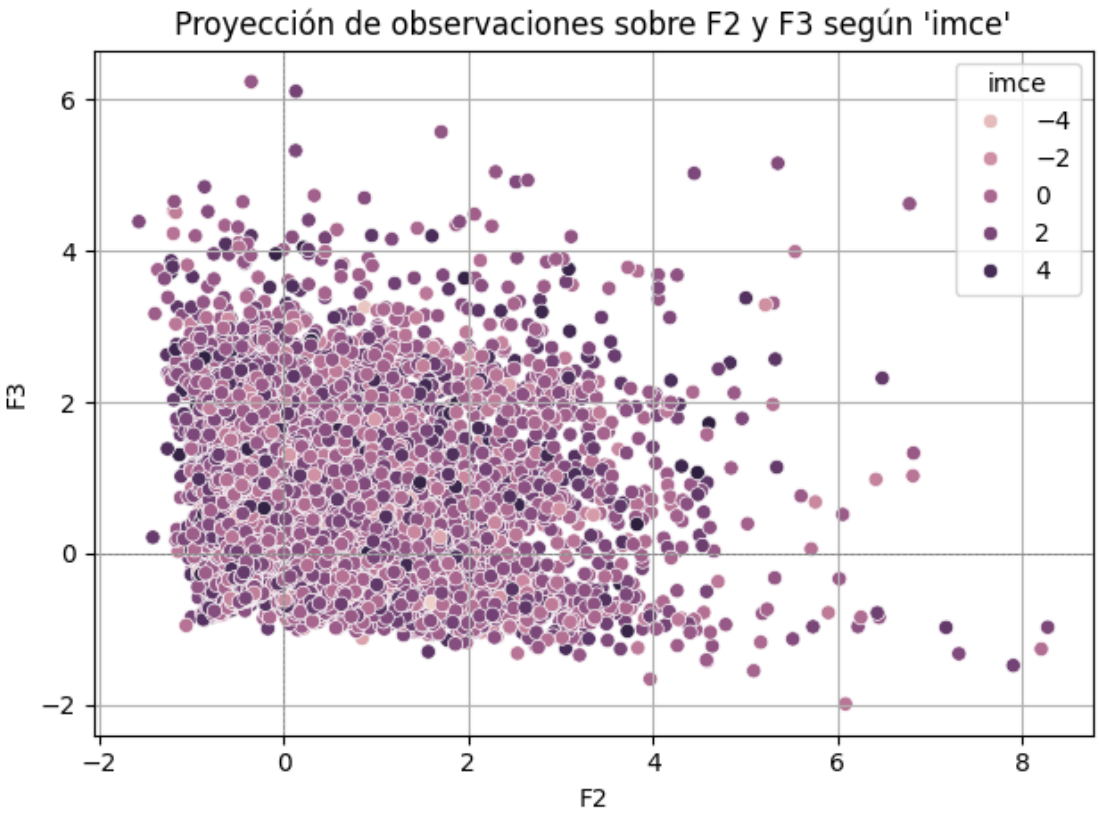

Proyección de observaciones sobre F1 y F2 según 'edad'
edad
64
72
80
88
96
104
F1
F2
−2
−1
0
1
2
3
4
5
6
8

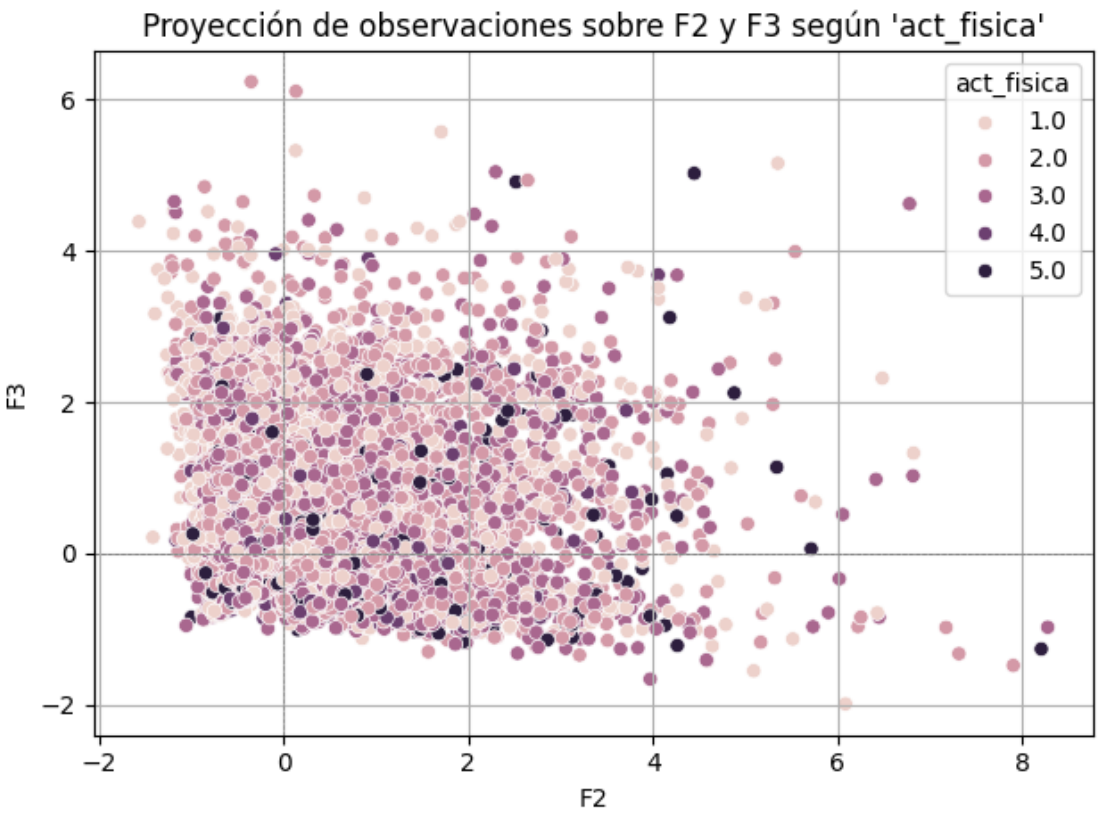

Proyección de observaciones sobre F1 y F2 según 'imce'
imce
−4
−2
0
2
4
F2
F1
8
6
4
2
0
−2
−2
−1
0
1
2
3
4
5
6


Proyección de observaciones sobre F1 y F2 según 'act_fisica'
act_fisica
1.0
2.0
3.0
4.0
5.0
F1
F2
−2
−1
0
1
2
3
4
5
6
8


Proyección de observaciones sobre F2 y F3 según 'sexo'
sexo
0
1
F3
F2
6
4
2
0
−2
−2
0
2
4
6
8


Proyección de observaciones sobre F2 y F3 según 'edad'
edad
64
72
80
88
96
104
F3
F2
6
4
2
0
−2
−2
0
2
4
6
8


Proyección de observaciones sobre F2 y F3 según 'imce'
imce
−4
−2
0
2
4
F3
F2
6
4
2
0
−2
−2
0
2
4
6
8



9 RESPUESTA PREGUNTA 3:

NO HAY PATRONES DETECTADOS PARA NINGUN FACTOR NI PARA NINGUNA PARTICION POR GRUPOS DE UNA CATEGORIA(HUE), ESTO SE PUEDE DEBER A QUE EN EL ANALISIS PCA HAY VARIABLES QUE SE RELACIONAN BIEN CON UN FACTOR PERO NO TAN MAL CON OTRO, POR ELLO QUE EN GENERAL PODEMOS VER QUE LA DATA DE DiSTRIBUYE COMO UNA NUBE "UNIFORME" SIN MOSTRAR ESTRUCTURAS DEFINIDAS EN PARTICULAR PARA ALGÚN CASO.

ADEMÁS PUEDE QUE EN VERDAD NO HAYAN DIFERENCIAS O SEGMENTACIONES SUSTANCIALES PARA AQUELLAS VARIABLES GRAFICADAS, O BIEN SU DIFERENCIA ES MINIMA Y PRACTICAMENTE A NIVEL VISUAL NO SE PUEDEN DETECTAR PATRONES SIGNIFICATIVOS.

	F1	F2	F3
sk1	0.171221	0.540217	0.151458
sk2	0.174149	0.512531	0.354330
sk3	0.179785	0.587108	0.161938
sk4	0.195805	0.685256	0.194526
sk5	0.121301	0.214133	0.716727
sk6	0.162524	0.203313	0.486711

```
sk7   0.070301  0.074530  0.146613
sk8   0.269211  0.097481  0.478541
sk9   0.495887  0.205273  0.201636
sk10  0.555989  0.121166  0.127875
sk11  0.665920  0.206954  0.202009
sk12  0.524127  0.109526  0.141501
sk13  0.480548  0.151480  0.131725

array([[0.171221, 0.540217, 0.151458],
       [0.174149, 0.512531, 0.354330],
       [0.179785, 0.587108, 0.161938],
       [0.195805, 0.685256, 0.194526],
       [0.121301, 0.214133, 0.716727],
       [0.162524, 0.203313, 0.486711],
       [0.070301, 0.074530, 0.146613],
       [0.269211, 0.097481, 0.478541],
       [0.495887, 0.205273, 0.201636],
       [0.555989, 0.121166, 0.127875],
       [0.665920, 0.206954, 0.202009],
       [0.524127, 0.109526, 0.141501],
       [0.480548, 0.151480, 0.131725]])
```

10 PREGUNTA 4

RECORDEMOS: Correlacion - sk1 -> Factor 2 alta - sk2 -> Factor 2 alta - sk3 -> Factor 2 alta - sk4 -> Factor 2 alta(llega de 60 a 80%) - sk5 -> Factor 3 alta - sk6 -> Factor 3 alta - sk7 -> Factor 3 baja(por "descarte se fue al que menos lo"rechaza") - sk8 -> Factor 3 alta - sk9 -> Factor 1 alta(+60%) - sk10 -> Factor 1 alta - sk11 -> Factor 1 alta - sk12 -> Factor 1 alta - sk13 -> Factor 1 alta

Respuesta pregunta 4 1/3

NOMBRAMOS A LOS FACTORES DE LAS VARIABLES:

NOMBRE: EXPRESIVIDAD (QUE TANTO EXPRESA EMOCIONES)

- sk1: muestra afecto a padres (1: siempre - 5: nunca)
- sk2: muestra afecto a sus pares (1: siempre - 5: nunca)
- sk3: expresa sus sentimientos (1: siempre - 5: nunca)
- sk4: usa gestos para mostrar sentimientos (1: siempre - 5: nunca)

NOMBRE: Interactividad

- sk5: juega con otros (1: siempre - 5: nunca)
- sk6: comparte sus cosas con otros (1: siempre - 5: nunca)
- sk7: es agresivo (1: siempre - 5: nunca)
- sk8: participa en juegos grupales (1: siempre - 5: nunca)

NOMBRE: INTERES (QUE TAN INTERSADO ES EN COSAS EN GENERAL)

- sk9: hace preguntas a adultos (1: siempre - 5: nunca)
- sk10: tiene interes por libros (1: siempre - 5: nunca)
- sk11: tiene interes por su entorno (1: siempre - 5: nunca)
- sk12: juega a armar y desarmar cosas (1: siempre - 5: nunca)
- sk13: tiene expresiones artisticas (1: siempre - 5: nunca)

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.170
Number of iterations: 41
Params: 1.350 1.249 1.098 1.274 1.875 1.637 1.837 1.057 0.553 1.169 0.049 0.156
0.057 0.151 0.051 0.091 0.095 0.595 0.200 0.442 0.713 0.241 0.200 0.159 0.156
0.366 1.415 0.472 0.278

       DoF  DoF Baseline         chi2  chi2 p-value  chi2 Baseline       CFI  \
Value   62            78  5047.485143           0.0   82399.270173  0.939439

            GFI      AGFI       NFI      TLI     RMSEA        AIC         BIC  \
Value  0.938744  0.922935  0.938744  0.92381  0.052037  57.660056  298.324318

         LogLik
Value  0.169972

              lval  op            rval  Estimate  Est. Std  Std. Err  \
0              sk9   ~         INTERES  1.000000  0.594109         -
1             sk10   ~         INTERES  1.350324  0.563032  0.018557
2             sk11   ~         INTERES  1.249408  0.736260  0.014749
3             sk12   ~         INTERES  1.097547  0.540387  0.015532
4             sk13   ~         INTERES  1.274438  0.506283  0.018927
5              sk1   ~    EXPRESIVIDAD  1.000000  0.584288         -
6              sk2   ~    EXPRESIVIDAD  1.875247  0.645980  0.023693
7              sk3   ~    EXPRESIVIDAD  1.637167  0.630289  0.020993
8              sk4   ~    EXPRESIVIDAD  1.836507  0.714421  0.022023
9              sk5   ~  INTERACTIVIDAD  1.000000  0.707058         -
10             sk6   ~  INTERACTIVIDAD  1.056561  0.567670  0.014605
11             sk7   ~  INTERACTIVIDAD  0.553107  0.180486  0.021165
12             sk8   ~  INTERACTIVIDAD  1.169148  0.557444  0.016355
13    EXPRESIVIDAD  ~~    EXPRESIVIDAD  0.049100  1.000000  0.001009
14  INTERACTIVIDAD  ~~  INTERACTIVIDAD  0.155736  1.000000    0.0028
15  INTERACTIVIDAD  ~~    EXPRESIVIDAD  0.057060  0.652526  0.000991
16         INTERES  ~~         INTERES  0.151397  1.000000  0.003066
17         INTERES  ~~    EXPRESIVIDAD  0.051363  0.595731  0.000965
```

18	INTERES	~~	INTERACTIVIDAD	0.090571	0.589845	0.001657
19	sk1	~~	sk1	0.094723	0.658607	0.000913
20	sk10	~~	sk10	0.594764	0.682995	0.005657
21	sk11	~~	sk11	0.199643	0.457921	0.002526
22	sk12	~~	sk12	0.442155	0.707982	0.004131
23	sk13	~~	sk13	0.713431	0.743678	0.006514
24	sk2	~~	sk2	0.241110	0.582710	0.002496
25	sk3	~~	sk3	0.199673	0.602736	0.002024
26	sk4	~~	sk4	0.158857	0.489603	0.001869
27	sk5	~~	sk5	0.155779	0.500069	0.002139
28	sk6	~~	sk6	0.365641	0.677751	0.003703
29	sk7	~~	sk7	1.414940	0.967425	0.011763
30	sk8	~~	sk8	0.472177	0.689256	0.004721
31	sk9	~~	sk9	0.277532	0.647035	0.002718

	z-value	p-value
0	-	-
1	72.76701	0.0
2	84.708688	0.0
3	70.665475	0.0
4	67.333174	0.0
5	-	-
6	79.148102	0.0
7	77.985509	0.0
8	83.389031	0.0
9	-	-
10	72.341318	0.0
11	26.132993	0.0
12	71.487126	0.0
13	48.654517	0.0
14	55.619368	0.0
15	57.573455	0.0
16	49.372653	0.0
17	53.218047	0.0
18	54.672277	0.0
19	103.731497	0.0
20	105.132842	0.0
21	79.043435	0.0
22	107.036841	0.0
23	109.518269	0.0
24	96.607628	0.0
25	98.66952	0.0
26	85.004695	0.0
27	72.844592	0.0
28	98.752827	0.0
29	120.287739	0.0
30	100.009027	0.0

```
 31  102.112579     0.0


Matriz de correlaciones estandarizadas entre factores latentes:

                INTERES  EXPRESIVIDAD  INTERACTIVIDAD
INTERES           1.000         0.596           0.590
EXPRESIVIDAD      0.596         1.000           0.653
INTERACTIVIDAD    0.590         0.653           1.000
```

RESPUESTA PREGUNTA 4 2/3 REPORTE DE IMPORTANCIA

RECORDEMOS: EST STD - sk1 -> Factor 2 alta | - sk2 -> Factor 2 alta |
- sk3 -> Factor 2 alta |
- sk4 -> Factor 2 alta |(llega de 60 a 80%)

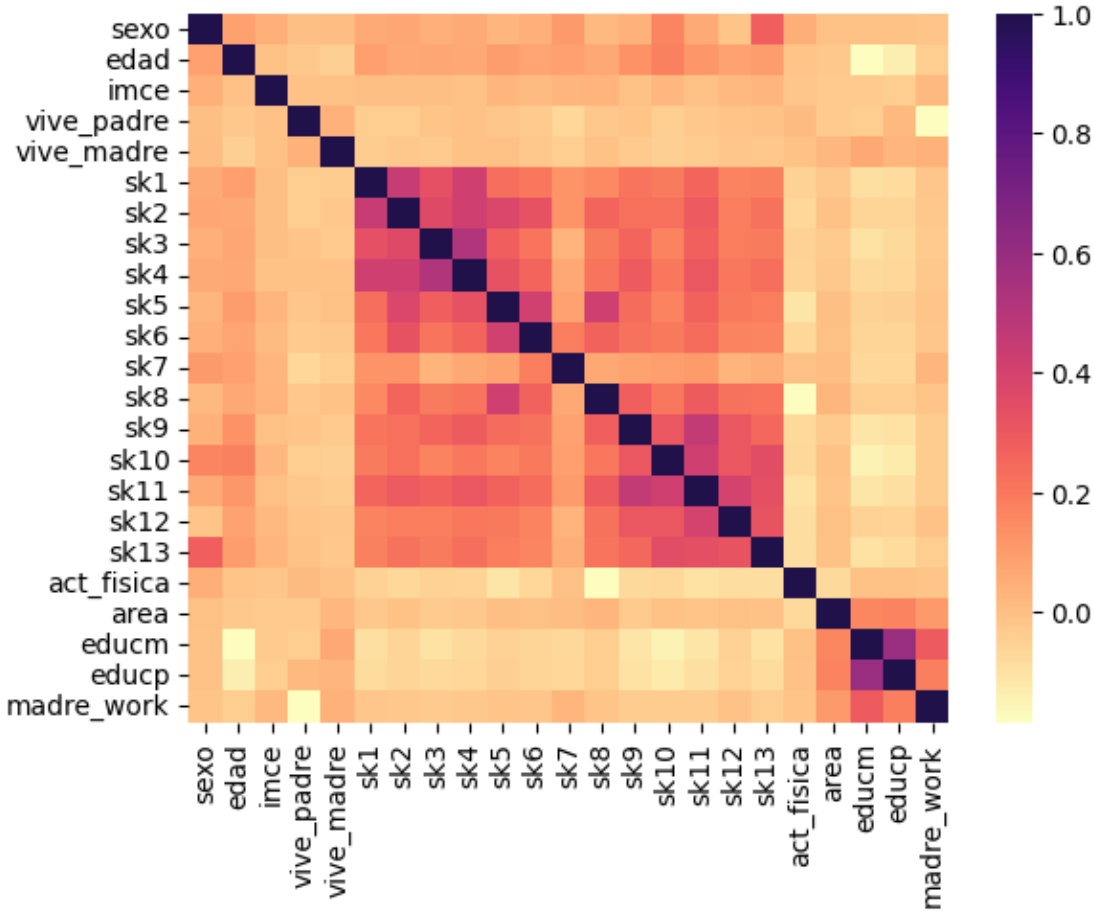
- sk5 -> Factor 3 alta
- sk6 -> Factor 3 alta
- sk7 -> Factor 3 baja(por "descarte se fue al que menos lo"rechaza")
- sk8 -> Factor 3 alta
- sk9 -> Factor 1 alta (+60%)
- sk10 -> Factor 1 alta
- sk11 -> Factor 1 alta
- sk12 -> Factor 1 alta
- sk13 -> Factor 1 alta

RESPUESTA 3/3 PREGUNTA 4:

DE LA TABLA ANTERIOR DE OBTIENE QUE:

HAY BASTANTE CORRELACION, PUEDE QUE EN VERDAD HAGA FALTA CONSIDERAR MAS VARIABLES(o usar otras) PARA DISCRIMINAR MEJOR LOS GRUPOS, ADEMAS PUEDE SIGNIFICAR QUE EL MODELO CAPTA UNA PARTE DE LOS DATOS DE UN FACTOR PARA OTRO, Y DADO QUE COMPARTEN ESO… PUES, LA CORRELACION.

```
<Axes: >
```



Notas propias

```
PARA EL FACTOR 1
sexo          0.134123
edad          0.161821
imce          0.014323
vive_padre   -0.025666
vive_madre   -0.043864
sk1           0.210225
sk2           0.213820
sk3           0.220740
sk4           0.240410
sk5           0.148933
sk6           0.199546
sk7           0.086316
sk8           0.330537
sk9           0.608850
sk10          0.682643
sk11          0.817615
```

```
sk12          0.643523
sk13          0.590016
act_fisica   -0.112168
area         -0.015308
educm        -0.139952
educp        -0.122090
madre_work   -0.042472
F1            1.000000
F2            0.150471
F3            0.155945
Name: F1, dtype: float64


PARA EL FACTOR 2
sexo          0.061338
edad          0.066742
imce         -0.010772
vive_padre   -0.027033
vive_madre   -0.025109
sk1           0.670739
sk2           0.636364
sk3           0.728958
sk4           0.850820
sk5           0.265870
sk6           0.252436
sk7           0.092537
sk8           0.121034
sk9           0.254869
sk10          0.150440
sk11          0.256956
sk12          0.135989
sk13          0.188079
act_fisica   -0.037446
area         -0.032149
educm        -0.090972
educp        -0.077175
madre_work   -0.027779
F1            0.150471
F2            1.000000
F3            0.201259
Name: F2, dtype: float64


PARA EL FACTOR 3
sexo          0.027797
edad          0.093482
imce          0.032183
vive_padre   -0.033047
```

```
vive_madre   -0.012232
sk1           0.193918
sk2           0.453663
sk3           0.207335
sk4           0.249059
sk5           0.917654
sk6           0.623155
sk7           0.187715
sk8           0.612696
sk9           0.258162
sk10          0.163724
sk11          0.258640
sk12          0.181170
sk13          0.168653
act_fisica   -0.134358
area          0.013768
educm        -0.047298
educp        -0.048830
madre_work   -0.012230
F1            0.155945
F2            0.201259
F3            1.000000
Name: F3, dtype: float64
```

```
sk11    0.817615
sk10    0.682643
sk12    0.643523
sk9     0.608850
sk13    0.590016
sk8     0.330537
sk4     0.240410
sk3     0.220740
sk2     0.213820
sk1     0.210225
sk6     0.199546
sk5     0.148933
sk7     0.086316
Name: F1, dtype: float64
```

NOTEMOS QUE PARA EL FACTOR 1 LO MAS ALTOS SON LOS QUE EN EFECTO SE ELIGIERON PARA PERTENECER A ESE FACTOR

```
sk4     0.850820
sk3     0.728958
sk1     0.670739
sk2     0.636364
sk5     0.265870
```

```
sk11    0.256956
sk9     0.254869
sk6     0.252436
sk13    0.188079
sk10    0.150440
sk12    0.135989
sk8     0.121034
sk7     0.092537
Name: F2, dtype: float64
```

```
sk5     0.917654
sk6     0.623155
sk8     0.612696
sk2     0.453663
sk11    0.258640
sk9     0.258162
sk4     0.249059
sk3     0.207335
sk1     0.193918
sk7     0.187715
sk12    0.181170
sk13    0.168653
sk10    0.163724
Name: F3, dtype: float64
```

fin notas propias

11 Pregunta 5

```
Requirement already satisfied: StepMix in
d:\benja\universidad\python(lenguaje)\lib\site-packages (2.2.1)
```

```
WARNING: Ignoring invalid distribution ~ip
(d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages)
WARNING: Ignoring invalid distribution ~ip
(d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages)
WARNING: Ignoring invalid distribution ~ip
(d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages)
```

```
Requirement already satisfied: numpy in
d:\benja\universidad\python(lenguaje)\lib\site-packages (from StepMix) (2.1.2)
Requirement already satisfied: pandas in
d:\benja\universidad\python(lenguaje)\lib\site-packages (from StepMix) (2.2.3)
Requirement already satisfied: scikit-learn>=1.0.0 in
d:\benja\universidad\python(lenguaje)\lib\site-packages (from StepMix) (1.6.1)
Requirement already satisfied: scipy in
d:\benja\universidad\python(lenguaje)\lib\site-packages (from StepMix) (1.15.2)
Requirement already satisfied: tqdm in
```

```
d:\benja\universidad\python(lenguaje)\lib\site-packages (from StepMix) (4.67.1)
Requirement already satisfied: joblib>=1.2.0 in
d:\benja\universidad\python(lenguaje)\lib\site-packages (from scikit-
learn>=1.0.0->StepMix) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in
d:\benja\universidad\python(lenguaje)\lib\site-packages (from scikit-
learn>=1.0.0->StepMix) (3.6.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
c:\users\benja\appdata\roaming\python\python313\site-packages (from
pandas->StepMix) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
d:\benja\universidad\python(lenguaje)\lib\site-packages (from pandas->StepMix)
(2025.2)
Requirement already satisfied: tzdata>=2022.7 in
d:\benja\universidad\python(lenguaje)\lib\site-packages (from pandas->StepMix)
(2025.2)
Requirement already satisfied: six>=1.5 in
c:\users\benja\appdata\roaming\python\python313\site-packages (from python-
dateutil>=2.8.2->pandas->StepMix) (1.16.0)
Requirement already satisfied: colorama in
c:\users\benja\appdata\roaming\python\python313\site-packages (from
tqdm->StepMix) (0.4.6)
```

```
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
```

```
Fitting StepMix…
```

```
Initializations (n_init) : 100%|      | 1/1 [00:00<00:00,  2.76it/s,
max_LL=-1.2e+6, max_avg_LL=-40.3]
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
```

```
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [00:00<00:00,  1.35it/s,
max_LL=-1.08e+6, max_avg_LL=-36.4]
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [00:02<00:00,  2.38s/it,
max_LL=-1.04e+6, max_avg_LL=-34.9]
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
```

```
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [00:02<00:00,  2.52s/it,
max_LL=-1.03e+6, max_avg_LL=-34.5]
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|     | 1/1 [00:06<00:00,  6.60s/it,
max_LL=-1.02e+6, max_avg_LL=-34.3]
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
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packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
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  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
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packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(

Fitting StepMix…
```

```
Initializations (n_init) : 100%|      | 1/1 [00:05<00:00,  5.55s/it,
max_LL=-1.01e+6, max_avg_LL=-33.9]
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
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d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
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renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:13<00:00, 13.32s/it,
max_LL=-9.88e+5, max_avg_LL=-33.3]
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
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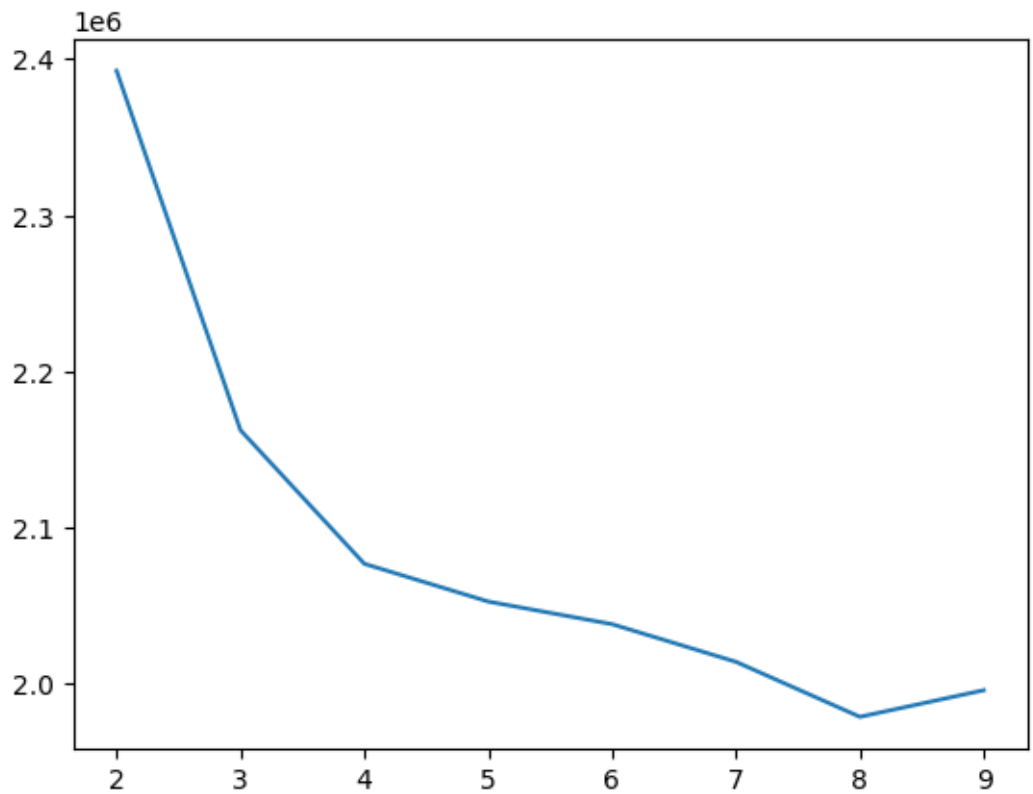
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:11<00:00, 11.51s/it,
max_LL=-9.96e+5, max_avg_LL=-33.5]
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
```

```
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
```

```
<Axes: >
```




```
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
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renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(
```

```
Fitting StepMix…
```

```
Initializations (n_init) : 100%|      | 1/1 [00:09<00:00,  9.03s/it,
```

```
max_LL=-5.13e+4, max_avg_LL=-1.73]
================================================================================
MODEL REPORT
================================================================================
    ==============================================================================
    Measurement model parameters
    ==============================================================================
          model_name           gaussian_diag
          class_no                         0       1       2
          param       variable
          covariances imce            0.9629  1.4229  2.7533
          means       imce            0.5012  1.8813 -0.0576
    ==============================================================================
    Class weights
    ==============================================================================
        Class 1 : 0.49
        Class 2 : 0.41
        Class 3 : 0.10
    ==============================================================================
    Fit for 3 latent classes
    ==============================================================================
    Estimation method             : 1-step
    Number of observations        : 29696
    Number of latent classes      : 3
    Number of estimated parameters: 8
    Log-likelihood (LL)           : -51339.8343
    -2LL                          : 102679.6686
    Average LL                    : -1.7288
    AIC                           : 102695.67
    BIC                           : 102762.06
    CAIC                          : 102770.06
    Sample-Size Adjusted BIC      : 102819.02
    Entropy                       : 22029.9267
    Scaled Relative Entropy       : 0.3247


d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\stepmix\stepmix.py:968:
ConvergenceWarning: Initializations did not converge. Try different init
parameters, or increase max_iter, abs_tol, rel_tol or check for degenerate data.
  warnings.warn(
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
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```

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packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
  warnings.warn(

       imce
1      0.71
3      2.05
4      1.05
5      1.39
6      2.75
…     …
31050  1.63
31051  2.57
31052  2.12
31053 -0.43
31054 -0.55

[29696 rows x 1 columns]

       sexo  edad  imce  vive_padre  vive_madre  sk1  sk2  sk3  sk4  sk5  …  \
1         0    76  0.71           0           1    1    1    1    1    1  …
3         1    84  2.05           1           1    1    1    1    1    1  …
4         0    86  1.05           1           1    1    1    1    1    1  …
5         0    74  1.39           1           1    1    2    1    1    1  …
6         1    91  2.75           1           1    1    1    1    2    2  …
…     …   …    …         …          …  …  …  …  …  …  …
31050     0    78  1.63           1           1    1    3    1    2    2  …
31051     1    79  2.57           1           1    1    1    2    2    1  …
31052     0    78  2.12           1           1    1    1    1    1    1  …
31053     1    78 -0.43           1           1    1    1    1    1    2  …
31054     0    78 -0.55           0           1    2    2    1    1    1  …

       sk9  sk10  sk11  sk12  sk13  act_fisica  area  educm  educp  madre_work
1        1     1     1     1     1         5.0     0    8.0      8           1
3        1     1     1     1     1         2.0     1   16.0     12          -1
4        1     1     1     1     1         1.0     1   17.0     15           0
5        1     1     1     1     1         4.0     0    8.0      8          -1
6        3     3     3     2     2         2.0     1   20.0     19           1
…    …   …   …   …   …         …   …    …    …         …
31050    2     2     2     1     1         2.0     1   13.0     13          -1
31051    1     3     2     1     4         3.0     1   18.0     19           0
31052    1     3     1     1     1         3.0     1   13.0      9           1
31053    1     2     1     1     2         2.0     1   17.0     15           1
```

```
31054    1     1     1     1     1         1.0     1   18.0     11           1

[29696 rows x 23 columns]
```

PROBAMOS ELIMINAR SK/

```
Index(['sexo', 'edad', 'imce', 'vive_padre', 'vive_madre', 'sk1', 'sk2', 'sk3',
       'sk4', 'sk5', 'sk6', 'sk8', 'sk9', 'sk10', 'sk11', 'sk12', 'sk13',
       'act_fisica', 'area', 'educm', 'educp', 'madre_work'],
      dtype='object')

       sk1  sk2  sk3  sk4  sk5  sk6  sk7  sk8  sk9  sk10  sk11  sk12  sk13
1        1    1    1    1    1    1    4    1    1     1     1     1     1
3        1    1    1    1    1    1    2    1    1     1     1     1     1
4        1    1    1    1    1    1    3    1    1     1     1     1     1
5        1    2    1    1    1    1    4    1    1     1     1     1     1
6        1    1    1    2    2    2    3    2    3     3     3     2     2
…    …   …   …   …   …   …   …   …   …   …   …   …   …
31050    1    3    1    2    2    1    1    3    2     2     2     1     1
31051    1    1    2    2    1    1    1    2    1     3     2     1     4
31052    1    1    1    1    1    1    1    1    1     3     1     1     1
31053    1    1    1    1    2    2    1    2    1     2     1     1     2
31054    2    2    1    1    1    1    5    1    1     1     1     1     1

[29696 rows x 13 columns]

d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
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  warnings.warn(

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:04<00:00,  4.16s/it,
max_LL=-5.14e+4, max_avg_LL=-1.73]

==============================================================================
MODEL REPORT
==============================================================================
    ==========================================================================
    Measurement model parameters
    ==========================================================================
          model_name     gaussian_unit
          class_no                   0       1       2
          param variable
```

```
      means imce            -1.949  2.4056  0.5984
    ==============================================================================
    Class weights
    ==============================================================================
        Class 1 : 0.02
        Class 2 : 0.27
        Class 3 : 0.71
    ==============================================================================
    Fit for 3 latent classes
    ==============================================================================
    Estimation method             : 1-step
    Number of observations        : 29696
    Number of latent classes      : 3
    Number of estimated parameters: 5
    Log-likelihood (LL)           : -51364.3251
    -2LL                          : 102728.6503
    Average LL                    : -1.7297
    AIC                           : 102738.65
    BIC                           : 102780.14
    CAIC                          : 102785.14
    Sample-Size Adjusted BIC      : 102815.75
    Entropy                       : 11499.4902
    Scaled Relative Entropy       : 0.6475


d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-packages\sklearn\base.py:474:
FutureWarning: `BaseEstimator._validate_data` is deprecated in 1.6 and will be
removed in 1.7. Use `sklearn.utils.validation.validate_data` instead. This
function becomes public and is part of the scikit-learn developer API.
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```

```
d:\benja\Universidad\PYTHON(LENGUAJE)\Lib\site-
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packages\sklearn\utils\deprecation.py:151: FutureWarning: 'force_all_finite' was
renamed to 'ensure_all_finite' in 1.6 and will be removed in 1.8.
```

```
  warnings.warn(
C:\Users\benja\AppData\Local\Temp\ipykernel_4144\2945546199.py:8:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df2['pred']=model.predict(df_solo)

       imce  pred
1      0.71     2
3      2.05     1
4      1.05     2
5      1.39     2
6      2.75     1
…     …   …
31050  1.63     2
31051  2.57     1
31052  2.12     1
31053 -0.43     2
31054 -0.55     2

[29696 rows x 2 columns]

       imce  pred
1      0.71     2
3      2.05     1
4      1.05     2
5      1.39     2
6      2.75     1
…     …   …
31050  1.63     2
31051  2.57     1
31052  2.12     1
31053 -0.43     2
31054 -0.55     2

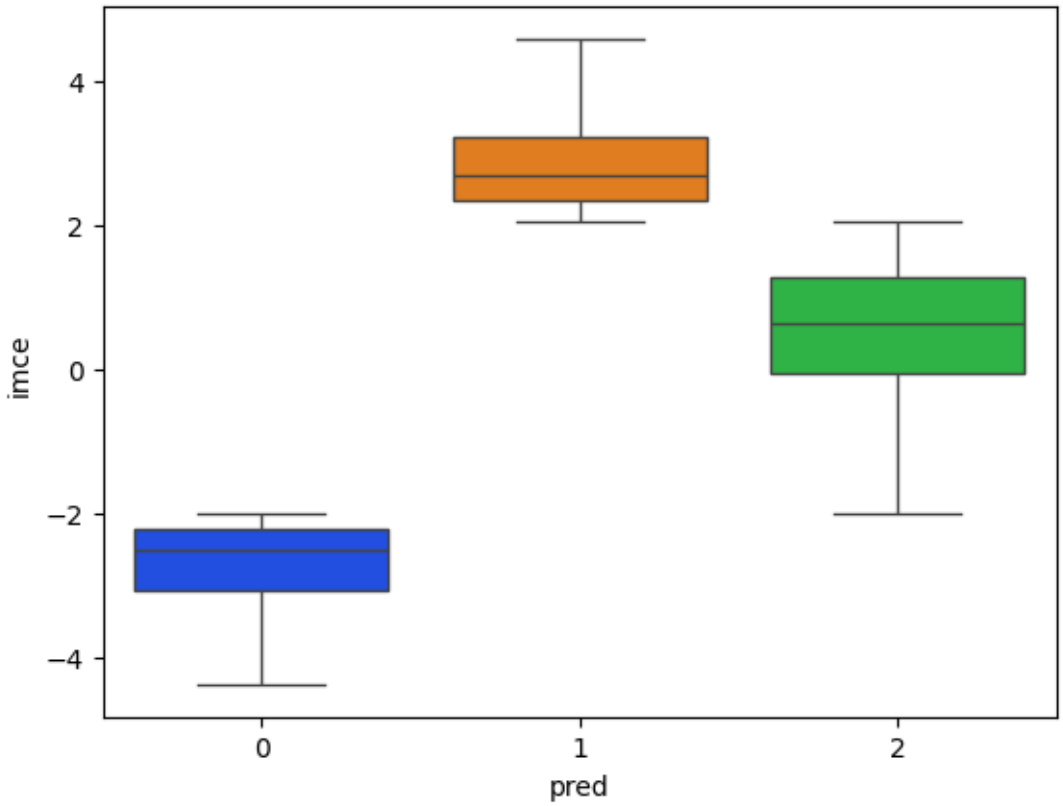
[29696 rows x 2 columns]

C:\Users\benja\AppData\Local\Temp\ipykernel_4144\1630985841.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.boxplot(data=df2, x='pred',y="imce",palette="bright", showfliers=False)
```

```
<Axes: xlabel='pred', ylabel='imce'>
```



12 Respuesta pregunta 5

SE PUEDE CONCLUIR QUE LA SEPARACION QUE SE HIZO FUE PRODUCTO DE ENCONRAR 3 GRUPOS DE PERSONAS, EL AZUL, PERSONAS CON UN IMCE MUY BAJO, TENDIENDO A DESNUTRICION, OTRO, VERDE, DONDE ESTA LA MAYORIA DE LOS INDIVIDUOS, EN UN RANGO "NORMAL", Y OTRO, NARANJO, QUE HABLA DEL GRUPO QUE TIENE SOBREPESO EN EXCESO

NOTAR QUE EL GRUPO VERDE IGUAL TIENE ALGUNAS PERSONAS DE IMCE ALGO ALTO, PERO CLARO…EL GRUPO NARANJO ES CLARAMENTE UN EXTREMO

13 Pregunta 6

MODELO SEM BASADO EN CFA

=~ **define factores latentes** relaciona

```
Index(['sexo', 'edad', 'imce', 'vive_padre', 'vive_madre', 'sk1', 'sk2', 'sk3',
       'sk4', 'sk5', 'sk6', 'sk7', 'sk8', 'sk9', 'sk10', 'sk11', 'sk12',
```

```
       'sk13', 'act_fisica', 'area', 'educm', 'educp', 'madre_work'],
      dtype='object')

       imce  pred
1      0.71     2
3      2.05     1
4      1.05     2
5      1.39     2
6      2.75     1
…     …   …
31050  1.63     2
31051  2.57     1
31052  2.12     1
31053 -0.43     2
31054 -0.55     2

[29696 rows x 2 columns]

Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.196
Number of iterations: 53
Params: 1.350 1.249 1.096 1.273 1.880 1.624 1.825 0.824 1.029 1.132 0.004 -0.275
0.226 -0.019 -0.115 -0.029 0.089 0.152 0.052 0.049 0.159 0.058 1.871 0.095 0.595
0.200 0.442 0.714 0.240 0.201 0.160 0.151 0.371 1.430 0.473 0.278
             lval  op            rval  Estimate  Std. Err     z-value  \
0             sk9   ~         INTERES  1.000000         -           -
1            sk10   ~         INTERES  1.349542  0.018552   72.745314
2            sk11   ~         INTERES  1.248745  0.014751   84.654497
3            sk12   ~         INTERES  1.095991  0.015523   70.606431
4            sk13   ~         INTERES  1.272632  0.018916   67.276369
5             sk1   ~    EXPRESIVIDAD  1.000000         -           -
6             sk2   ~    EXPRESIVIDAD  1.879749   0.02364   79.515123
7             sk3   ~    EXPRESIVIDAD  1.624402  0.020863   77.861303
8             sk4   ~    EXPRESIVIDAD  1.824771  0.021871   83.433566
9             sk7   ~    EXPRESIVIDAD  0.823738  0.036259   22.718395
10            sk5   ~  INTERACTIVIDAD  1.000000         -           -
11            sk6   ~  INTERACTIVIDAD  1.028931  0.014526   70.833332
12            sk8   ~  INTERACTIVIDAD  1.131815  0.016126   70.184176
13           imce   ~         INTERES  0.004376  0.035568     0.12303
14           imce   ~    EXPRESIVIDAD -0.275273   0.07114   -3.869466
15           imce   ~  INTERACTIVIDAD  0.225774   0.04098    5.509318
16           imce   ~      act_fisica -0.018653  0.007389   -2.524581
17         INTERES  ~~  INTERACTIVIDAD  0.089344  0.001651   54.104244
18         INTERES  ~~         INTERES  0.151578   0.00307    49.37188
19         INTERES  ~~    EXPRESIVIDAD  0.051746  0.000969   53.401707
```

20	EXPRESIVIDAD	~~	EXPRESIVIDAD	0.049264	0.00101	48.790104
21	INTERACTIVIDAD	~~	INTERACTIVIDAD	0.158842	0.002856	55.619421
22	INTERACTIVIDAD	~~	EXPRESIVIDAD	0.057597	0.000996	57.819197
23	act_fisica	~~	sk8	-0.115224	0.004623	-24.921751
24	act_fisica	~~	sk5	-0.028846	0.002928	-9.850135
25	imce	~~	imce	1.870886	0.015401	121.477273
26	sk1	~~	sk1	0.094521	0.000911	103.714497
27	sk10	~~	sk10	0.594678	0.005659	105.076874
28	sk11	~~	sk11	0.199521	0.002529	78.886156
29	sk12	~~	sk12	0.442296	0.004133	107.020266
30	sk13	~~	sk13	0.713582	0.006517	109.5035
31	sk2	~~	sk2	0.239740	0.002488	96.376197
32	sk3	~~	sk3	0.201245	0.002028	99.255689
33	sk4	~~	sk4	0.160491	0.001868	85.902402
34	sk5	~~	sk5	0.150864	0.002183	69.114837
35	sk6	~~	sk6	0.371379	0.003731	99.536234
36	sk7	~~	sk7	1.429547	0.011809	121.055865
37	sk8	~~	sk8	0.473410	0.004706	100.598889
38	sk9	~~	sk9	0.277517	0.00272	102.0237

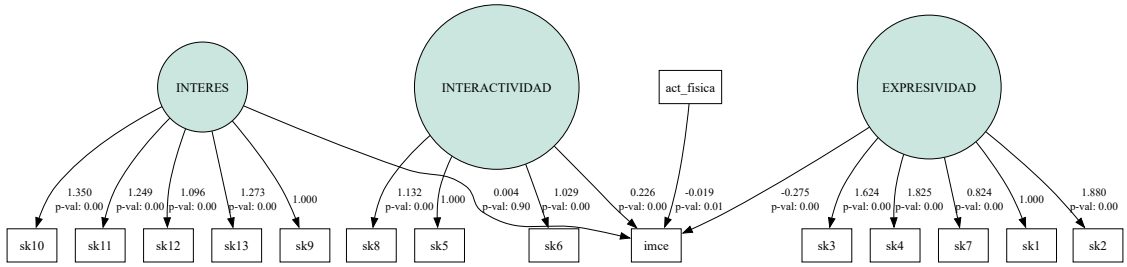
	p-value
0	-
1	0.0
2	0.0
3	0.0
4	0.0
5	-
6	0.0
7	0.0
8	0.0
9	0.0
10	-
11	0.0
12	0.0
13	0.902083
14	0.000109
15	0.0
16	0.011584
17	0.0
18	0.0
19	0.0
20	0.0
21	0.0
22	0.0
23	0.0
24	0.0
25	0.0
26	0.0

```
27       0.0
28       0.0
29       0.0
30       0.0
31       0.0
32       0.0
33       0.0
34       0.0
35       0.0
36       0.0
37       0.0
38       0.0
None

       DoF  DoF Baseline        chi2  chi2 p-value  chi2 Baseline       CFI  \
Value   84           106  5807.69965           0.0    83671.80356  0.931507

           GFI      AGFI      NFI       TLI     RMSEA        AIC         BIC  \
Value  0.93059  0.912411  0.93059  0.913568  0.047902  71.608856  370.364491

         LogLik
Value  0.195572
```




```
       DoF  DoF Baseline        chi2  chi2 p-value  chi2 Baseline       CFI  \
Value   84           106  5807.69965           0.0    83671.80356  0.931507

           GFI      AGFI      NFI       TLI     RMSEA        AIC         BIC  \
Value  0.93059  0.912411  0.93059  0.913568  0.047902  71.608856  370.364491

         LogLik
Value  0.195572
```


14 respuesta pregunta 6

DE LAS METRICAS NOTEMOS QUE EL TLI CFI RMSEA:

TLI $> 90\%$

RMSEA < 0.5%

CFI = 0.93 <0.95 (no es lo ideal) pero esta bastante bien

al considerar el imce como objeto de variacion el ajuste a los grupos es mucho mejor, ya que para las demas variables si bien hay diferencias, estas son minimas con la evidente diferencia que hay de los individuos en cuanto a su indice de masa corporal. Mas se decidio agregar la actividad fisica pues debe ser clave para su salud fisica, ademas, si bien puede jugar, esto habla de la frecuencia y o "intensidad" de sus actividades a lo largo del tiempo. mas de descartaron otras variables como la edad o el sexo por no ser relevantes a la hora de segmentar, puesto que para la edad la variabilidad es muy baja, y para sexo se vio antes que esta no afectaba a este grupo etario en mayor medida. respecto a los padres, se considero mas pertinente su actuar mas que esten, y puesto que se indique su interaccion indirectamente habla de su presencia, es decir, se cree mas relevante la interacción.

15 PREGUNTA 7

	sexo	edad	imce	vive_padre	vive_madre	sk1	sk2	sk3	sk4	sk5	…	\
39	1	84	-2.37	1	1	1	3	1	1	1	…	
108	0	105	-3.30	1	1	2	2	2	2	2	…	
163	1	105	-3.40	1	1	1	1	1	1	2	…	
243	0	102	-3.80	0	1	1	2	2	2	2	…	
451	1	98	-2.51	0	1	1	1	1	1	1	…	
…	…	…	…	…	…	…	…	…	…	…	…	
30544	0	80	-2.82	1	1	1	1	1	1	1	…	
30711	0	79	-3.39	1	1	1	1	1	1	2	…	
30742	1	79	-2.13	0	1	2	2	1	1	2	…	
30819	1	80	-2.36	1	1	1	2	1	1	1	…	
30829	1	79	-3.13	0	1	2	2	2	2	1	…	

	sk10	sk11	sk12	sk13	act_fisica	area	educm	educp	madre_work	\
39	3	3	1	5	5.0	1	6.0	9	-1	
108	2	2	2	2	2.0	1	8.0	12	-1	
163	3	2	2	3	3.0	0	11.0	2	-1	
243	3	3	3	2	1.0	1	9.0	12	0	
451	1	1	1	1	3.0	1	9.0	9	1	
…	…	…	…	…	…	…	…	…	…	
30544	1	1	1	3	5.0	1	13.0	13	-1	
30711	2	1	1	1	4.0	1	13.0	13	-1	
30742	3	1	2	1	3.0	1	13.0	13	-1	
30819	3	1	1	3	2.0	1	12.0	11	-1	
30829	2	1	1	2	2.0	1	13.0	11	1	

	pred
39	0
108	0
163	0
243	0
451	0

```
…      …
30544     0
30711     0
30742     0
30819     0
30829     0

[414 rows x 24 columns]

       sexo  edad  imce  vive_padre  vive_madre  sk1  sk2  sk3  sk4  sk5  …  \
3         1    84  2.05           1           1    1    1    1    1    1  …
6         1    91  2.75           1           1    1    1    1    2    2  …
17        1    74  2.53           1           1    1    2    1    1    1  …
24        0    85  3.41           0           1    2    3    1    2    2  …
25        1    84  2.98           1           1    1    1    1    1    1  …
…     …   …   …         …          …  …  …  …  …  …  …
31034     0    79  3.04           1           1    1    1    3    1    2  …
31035     0    80  3.22           1           1    1    1    2    1    3  …
31045     0    78  2.82           0           1    1    1    1    1    1  …
31051     1    79  2.57           1           1    1    1    2    2    1  …
31052     0    78  2.12           1           1    1    1    1    1    1  …

       sk10  sk11  sk12  sk13  act_fisica  area  educm  educp  madre_work  \
3         1     1     1     1         2.0     1   16.0     12          -1
6         3     3     2     2         2.0     1   20.0     19           1
17        1     1     1     1         2.0     1   13.0     13           1
24        2     2     2     2         2.0     0   13.0     13           0
25        1     1     1     4         3.0     0    9.0      9          -1
…     …   …   …   …         …   …    …    …         …
31034     2     1     1     1         4.0     1   13.0     13           1
31035     3     1     2     1         2.0     1   13.0     13           1
31045     2     1     2     1         3.0     1   19.0     19           1
31051     3     2     1     4         3.0     1   18.0     19           0
31052     3     1     1     1         3.0     1   13.0      9           1

       pred
3         1
6         1
17        1
24        1
25        1
…     …
31034     1
31035     1
31045     1
31051     1
31052     1
```

```
[6679 rows x 24 columns]

       sexo  edad  imce  vive_padre  vive_madre  sk1  sk2  sk3  sk4  sk5  …  \
1         0    76  0.71           0           1    1    1    1    1    1  …
4         0    86  1.05           1           1    1    1    1    1    1  …
5         0    74  1.39           1           1    1    2    1    1    1  …
7         1    92  1.53           1           1    1    3    1    1    3  …
8         1    69  1.96           1           1    1    2    3    2    1  …
…     …   …   …         …          …  …  …  …  …  …  …
31048     1    78  1.58           1           1    1    1    1    1    1  …
31049     1    79  1.73           1           0    1    1    2    1    1  …
31050     0    78  1.63           1           1    1    3    1    2    2  …
31053     1    78 -0.43           1           1    1    1    1    1    2  …
31054     0    78 -0.55           0           1    2    2    1    1    1  …

       sk10  sk11  sk12  sk13  act_fisica  area  educm  educp  madre_work  \
1         1     1     1     1         5.0     0    8.0      8           1
4         1     1     1     1         1.0     1   17.0     15           0
5         1     1     1     1         4.0     0    8.0      8          -1
7         4     3     2     4         1.0     1   13.0     11           0
8         4     2     2     3         2.0     0   10.0      9           1
…     …   …   …   …         …   …    …    …         …
31048     4     3     3     5         1.0     0   13.0     12          -1
31049     1     2     1     2         2.0     1   17.0     17           1
31050     2     2     1     1         2.0     1   13.0     13          -1
31053     2     1     1     2         2.0     1   17.0     15           1
31054     1     1     1     1         1.0     1   18.0     11           1

       pred
1         2
4         2
5         2
7         2
8         2
…     …
31048     2
31049     2
31050     2
31053     2
31054     2

[22603 rows x 24 columns]
```

modelo para grupo 0:

```
Name of objective: MLW
Optimization method: SLSQP
```

```
Optimization successful.
Optimization terminated successfully
Objective value: 0.512
Number of iterations: 54
Params: 1.055 1.117 0.827 1.077 2.466 2.187 2.759 1.134 1.093 1.253 -0.121 1.101
-0.449 0.080 -0.155 -0.000 0.091 0.223 0.047 0.028 0.152 0.048 0.469 0.081 0.653
0.215 0.350 0.762 0.240 0.199 0.140 0.165 0.407 1.496 0.541 0.281
              lval  op            rval  Estimate  Std. Err    z-value  \
0              sk9   ~         INTERES  1.000000         -          -
1             sk10   ~         INTERES  1.055059  0.120338   8.767439
2             sk11   ~         INTERES  1.117027  0.099458  11.231107
3             sk12   ~         INTERES  0.827340  0.090557   9.136115
4             sk13   ~         INTERES  1.077198  0.127243   8.465651
5              sk1   ~    EXPRESIVIDAD  1.000000         -          -
6              sk2   ~    EXPRESIVIDAD  2.465914  0.287997   8.562293
7              sk3   ~    EXPRESIVIDAD  2.187484  0.257542   8.493714
8              sk4   ~    EXPRESIVIDAD  2.758910  0.298773   9.234135
9              sk7   ~    EXPRESIVIDAD  1.133725  0.415145   2.730911
10             sk5   ~  INTERACTIVIDAD  1.000000         -          -
11             sk6   ~  INTERACTIVIDAD  1.093260  0.130107   8.402746
12             sk8   ~  INTERACTIVIDAD  1.252606  0.147852   8.472053
13            imce   ~         INTERES -0.120508   0.11847  -1.017198
14            imce   ~    EXPRESIVIDAD  1.100846  0.535128   2.057164
15            imce   ~  INTERACTIVIDAD -0.448753  0.218925  -2.049806
16            imce   ~      act_fisica  0.080053  0.032151   2.489866
17         INTERES  ~~  INTERACTIVIDAD  0.091173  0.015324   5.949665
18         INTERES  ~~         INTERES  0.223262  0.033185   6.727775
19         INTERES  ~~    EXPRESIVIDAD  0.047470   0.00758   6.262188
20    EXPRESIVIDAD  ~~    EXPRESIVIDAD  0.027929  0.005623   4.967307
21  INTERACTIVIDAD  ~~  INTERACTIVIDAD  0.151831   0.02365   6.420052
22  INTERACTIVIDAD  ~~    EXPRESIVIDAD  0.047831  0.007106   6.730753
23      act_fisica  ~~             sk8 -0.155495  0.040453  -3.843803
24      act_fisica  ~~             sk5 -0.000384  0.024571  -0.015642
25            imce  ~~            imce  0.469485   0.03477  13.502404
26             sk1  ~~             sk1  0.080632  0.006123   13.16792
27            sk10  ~~            sk10  0.653471  0.050781    12.8684
28            sk11  ~~            sk11  0.215475   0.02354   9.153477
29            sk12  ~~            sk12  0.349614  0.027668  12.635827
30            sk13  ~~            sk13  0.761975  0.058467  13.032563
31             sk2  ~~             sk2  0.240161  0.020275  11.845257
32             sk3  ~~             sk3  0.199300  0.016633  11.982013
33             sk4  ~~             sk4  0.139827  0.015701   8.905678
34             sk5  ~~             sk5  0.165131  0.018369   8.989819
35             sk6  ~~             sk6  0.406715  0.034316   11.85205
36             sk7  ~~             sk7  1.495581  0.104554  14.304384
37             sk8  ~~             sk8  0.540625  0.045482  11.886663
38             sk9  ~~             sk9  0.280706  0.025211  11.134335
```

```
     p-value
0          -
1        0.0
2        0.0
3        0.0
4        0.0
5          -
6        0.0
7        0.0
8        0.0
9   0.006316
10         -
11       0.0
12       0.0
13  0.309059
14   0.03967
15  0.040383
16  0.012779
17       0.0
18       0.0
19       0.0
20  0.000001
21       0.0
22       0.0
23  0.000121
24   0.98752
25       0.0
26       0.0
27       0.0
28       0.0
29       0.0
30       0.0
31       0.0
32       0.0
33       0.0
34       0.0
35       0.0
36       0.0
37       0.0
38       0.0
None

        DoF  DoF Baseline        chi2  chi2 p-value  chi2 Baseline       CFI  \
Value    84           106  212.133768  4.659606e-13    1370.658867  0.898681

             GFI      AGFI       NFI       TLI     RMSEA        AIC  \
Value  0.845232  0.804698  0.845232  0.872145  0.060774  70.975199
```

```
               BIC  LogLik
 Value  215.906374  0.5124
```

modelo para el grupo 1:

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.221
Number of iterations: 40
Params: 1.365 1.244 1.142 1.360 1.811 1.464 1.628 0.855 0.985 1.075 0.020 -0.132
0.135 0.003 -0.121 -0.032 0.097 0.156 0.055 0.058 0.184 0.065 0.453 0.096 0.625
0.210 0.470 0.742 0.243 0.218 0.163 0.164 0.396 1.478 0.529 0.273
              lval  op           rval  Estimate  Std. Err    z-value  \
0              sk9   ~         INTERES  1.000000         -          -
1             sk10   ~         INTERES  1.364937  0.039081  34.925519
2             sk11   ~         INTERES  1.244434  0.030567  40.712069
3             sk12   ~         INTERES  1.141896  0.033278  34.313474
4             sk13   ~         INTERES  1.360070  0.040764  33.364838
5              sk1   ~    EXPRESIVIDAD  1.000000         -          -
6              sk2   ~    EXPRESIVIDAD  1.811222   0.04611  39.280764
7              sk3   ~    EXPRESIVIDAD  1.464254  0.039637  36.941629
8              sk4   ~    EXPRESIVIDAD  1.627868   0.04031  40.383515
9              sk7   ~    EXPRESIVIDAD  0.855216  0.072208  11.843794
10             sk5   ~  INTERACTIVIDAD  1.000000         -          -
11             sk6   ~  INTERACTIVIDAD  0.984902  0.029422   33.47541
12             sk8   ~  INTERACTIVIDAD  1.074834   0.03287  32.699392
13            imce   ~         INTERES  0.019570  0.035681   0.548476
14            imce   ~    EXPRESIVIDAD -0.131803  0.065413  -2.014931
15            imce   ~  INTERACTIVIDAD  0.134842  0.038536   3.499126
16            imce   ~      act_fisica  0.003436  0.007922    0.43374
17         INTERES  ~~  INTERACTIVIDAD  0.096897  0.003745  25.876138
18         INTERES  ~~         INTERES  0.156265  0.006543  23.882934
19         INTERES  ~~    EXPRESIVIDAD  0.055036   0.00216  25.474148
20    EXPRESIVIDAD  ~~    EXPRESIVIDAD  0.057625  0.002367  24.345696
21  INTERACTIVIDAD  ~~  INTERACTIVIDAD  0.184137  0.006914  26.632896
22  INTERACTIVIDAD  ~~    EXPRESIVIDAD  0.065193  0.002359  27.636191
23      act_fisica  ~~             sk8 -0.121132  0.009927 -12.202508
24      act_fisica  ~~             sk5 -0.031956  0.006317  -5.058433
25            imce  ~~            imce  0.452836  0.007871  57.531054
26             sk1  ~~             sk1  0.095831  0.002013  47.614225
27            sk10  ~~            sk10  0.625189  0.012527  49.908461
28            sk11  ~~            sk11  0.210363  0.005503  38.226286
29            sk12  ~~            sk12  0.469681  0.009308   50.45921
30            sk13  ~~            sk13  0.741510  0.014478  51.217278
31             sk2  ~~             sk2  0.242824  0.005452  44.539204
```

32	sk3	~~	sk3	0.217877	0.00452	48.206906
33	sk4	~~	sk4	0.162731	0.003893	41.800994
34	sk5	~~	sk5	0.164308	0.005257	31.255623
35	sk6	~~	sk6	0.396473	0.008397	47.21391
36	sk7	~~	sk7	1.477618	0.02578	57.31727
37	sk8	~~	sk8	0.529453	0.010947	48.366885
38	sk9	~~	sk9	0.272884	0.00568	48.045945

	p-value
0	-
1	0.0
2	0.0
3	0.0
4	0.0
5	-
6	0.0
7	0.0
8	0.0
9	0.0
10	-
11	0.0
12	0.0
13	0.583365
14	0.043912
15	0.000467
16	0.664477
17	0.0
18	0.0
19	0.0
20	0.0
21	0.0
22	0.0
23	0.0
24	0.0
25	0.0
26	0.0
27	0.0
28	0.0
29	0.0
30	0.0
31	0.0
32	0.0
33	0.0
34	0.0
35	0.0
36	0.0
37	0.0
38	0.0

```
None

       DoF  DoF Baseline         chi2  chi2 p-value  chi2 Baseline       CFI  \
Value   84           106  1475.831645           0.0    19105.43338  0.926744

            GFI      AGFI       NFI       TLI     RMSEA        AIC  \
Value  0.922753  0.902522  0.922753  0.907557  0.049812  71.558068

              BIC    LogLik
Value  316.600116  0.220966
```

modelo para el grupo 2

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.191
Number of iterations: 44
Params: 1.348 1.253 1.088 1.247 1.893 1.670 1.877 0.799 1.043 1.148 -0.019
-0.052 0.017 -0.003 -0.112 -0.028 0.087 0.149 0.051 0.047 0.151 0.055 0.825
0.094 0.585 0.196 0.435 0.704 0.239 0.196 0.160 0.146 0.363 1.411 0.455 0.279
             lval  op            rval  Estimate  Std. Err     z-value  \
0             sk9   ~         INTERES  1.000000         -           -
1            sk10   ~         INTERES  1.347718  0.021349   63.128362
2            sk11   ~         INTERES  1.252889  0.017065   73.419018
3            sk12   ~         INTERES  1.088304  0.017813   61.096667
4            sk13   ~         INTERES  1.247394  0.021596   57.761252
5             sk1   ~    EXPRESIVIDAD  1.000000         -           -
6             sk2   ~    EXPRESIVIDAD  1.892808  0.027577   68.637677
7             sk3   ~    EXPRESIVIDAD  1.669939  0.024564   67.984533
8             sk4   ~    EXPRESIVIDAD  1.876888  0.025918   72.417086
9             sk7   ~    EXPRESIVIDAD  0.799406  0.042011   19.028389
10            sk5   ~  INTERACTIVIDAD  1.000000         -           -
11            sk6   ~  INTERACTIVIDAD  1.043230  0.016888   61.772113
12            sk8   ~  INTERACTIVIDAD  1.147543  0.018679   61.436527
13           imce   ~         INTERES -0.019237  0.027441   -0.701044
14           imce   ~    EXPRESIVIDAD -0.051741  0.055607   -0.930475
15           imce   ~  INTERACTIVIDAD  0.017427  0.032092    0.543019
16           imce   ~      act_fisica -0.003386  0.005567   -0.608149
17        INTERES  ~~  INTERACTIVIDAD  0.086863  0.001844   47.096035
18        INTERES  ~~         INTERES  0.149135  0.003491   42.720717
19        INTERES  ~~    EXPRESIVIDAD  0.050767  0.001092   46.472424
20   EXPRESIVIDAD  ~~    EXPRESIVIDAD  0.047383  0.001127   42.039615
21 INTERACTIVIDAD  ~~  INTERACTIVIDAD  0.151194  0.003129   48.326859
22 INTERACTIVIDAD  ~~    EXPRESIVIDAD  0.055473  0.001103   50.295222
23     act_fisica  ~~             sk8 -0.111922  0.005253  -21.306486
24     act_fisica  ~~             sk5 -0.027941  0.003325   -8.402549
25           imce  ~~            imce  0.824522  0.007757  106.296927
```

26	sk1	~~	sk1	0.094135	0.001033	91.155097
27	sk10	~~	sk10	0.584983	0.006383	91.649442
28	sk11	~~	sk11	0.195750	0.002865	68.326303
29	sk12	~~	sk12	0.435380	0.004656	93.507721
30	sk13	~~	sk13	0.703658	0.007334	95.948582
31	sk2	~~	sk2	0.238747	0.002818	84.716934
32	sk3	~~	sk3	0.196217	0.002285	85.888588
33	sk4	~~	sk4	0.159688	0.002144	74.47095
34	sk5	~~	sk5	0.146486	0.002397	61.099846
35	sk6	~~	sk6	0.363284	0.004185	86.816614
36	sk7	~~	sk7	1.411317	0.013355	105.67594
37	sk8	~~	sk8	0.455230	0.005208	87.407841
38	sk9	~~	sk9	0.278554	0.00312	89.288022

	p-value
0	-
1	0.0
2	0.0
3	0.0
4	0.0
5	-
6	0.0
7	0.0
8	0.0
9	0.0
10	-
11	0.0
12	0.0
13	0.483276
14	0.352125
15	0.587117
16	0.543089
17	0.0
18	0.0
19	0.0
20	0.0
21	0.0
22	0.0
23	0.0
24	0.0
25	0.0
26	0.0
27	0.0
28	0.0
29	0.0
30	0.0
31	0.0
32	0.0

```
33       0.0
34       0.0
35       0.0
36       0.0
37       0.0
38       0.0
None

       DoF  DoF Baseline        chi2  chi2 p-value  chi2 Baseline       CFI  \
Value   84           106  4312.07052           0.0   63405.212218  0.933205

            GFI     AGFI       NFI       TLI     RMSEA        AIC         BIC  \
Value  0.931992  0.91418  0.931992  0.915711  0.047191  71.618451  360.548617

         LogLik
Value  0.190774
```

16 Respuesta pregunta 7:

AL realizar el analisis pero esta vez en grupos, podemos decir para el caso del grupo...

grupo 0(desnutricion): no descrimina bien, no se parece al caso global de la pregunta 6

grupo 1(obesidad): descrimina bien, incluso parecido al caso global de la pregunta 6

grupo 2(rango normal): descrimina bien, incluso parecido al caso global de la pregunta 6

hay una diferencia en los casos del grupo 1 y 2, que es poca, mas el caso global tiene estadisticas ligeramente mayores, mas a priori, lo hicieron bien ambos, mas para el caso del grupo 0, esto no sucedio, tuvo un peor desempeño en todos los parametros considerablemente, bajando del 0.9 incluso, para los otros 2 su baja es apenas del 0.1-0.2 pero solo en el tli, por lo que puede indicar que el ajuste a los datos en este caso en un poco mas bajo.

estas diferencias principlamente se deberan a que los rangos de valores de las variables abarcan tanto al grupo 1 o 2, pero al grupo 0 lo dejan mas resagado, ademas que el valor del imce esta mucho mas alejado de cada uno de los otros grupos, no tan asi la situacion entre grupo 1 y 2